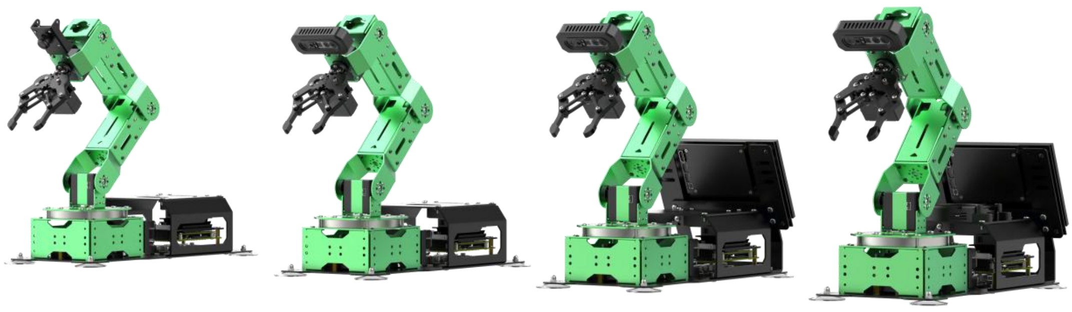
JetArm Introductory Manual V1.0



Hiwonder official website

Version	Released Date	Revised Content
V1.0	20231013	Initially released

Catalog

1. Introduction 6
1.1 Robot Version Instruction 7
1.1.1 JetArm Starter Kit 7
1.1.2 Standard Kit 7
1.1.3 Advanced Kit 8
1.1.4 Ultimate Kit 9
1.2 Packing List 9
1.2.1 Starter Kit Packing List 9
1.2.2 Standard Kit Packing List 10
1.2.3 Advanced Kit Packing List 10
1.2.4 Advanced Kit Packing List 11
2. Hardware Introduction 11
2.1 Hardware System 12
2.2 Electrical Control System Introduction 12
2.2.1 STM32 Controller 12
2.2.2 Power Instruction 14
2.3 Hardware Module Instruction 15
2.3.1 Main Controller 15
2.3.2 Robot Driver Board 16
2.3.3 Intelligent Bus Servo 17
2.3.4 Depth Camera 18

2.3.5 Microphone Module 21

2.3.6 Sound Card and Speaker 22

2.3.7 Screen 23

3. Accessory Assembly 24

3.1 Install 7-Inch LCD Screen 24

3.2 USB Interface Instruction 26

3.3 Install Suction Cups 26

4. Initial Startup and Test 27

4.1 Note 27

4.2 Connect Power Adapter 28

4.3 Robot Boot-up Status and Testing Instruction 28

4.3.1 Boot-up Status 28

4.3.2 Hardware Test 30

5. Installation and Connection 32

5.1 App Installation 32

5.1.1 APP Store Download 32

5.1.2 Hiwonder Official Website 32

5.2 Connection Mode Introduction 34

5.2.1 Direct Connection (Must Read) 34

5.2.2 LAN Mode（Optional） 38

6. Map Introduction and Device Placement 43

7. Position Calibration 48

8. APP Control 50

8.1 Preparation 51

8.2 APP Introduction 51

8.3 APP Operation 52

8.3.1Position Calibration 52

8.3.2 Robot Control 53

8.3.3 Object Sorting 54

8.3.4 Gesture Control 56

8.3.5 Target Tracking 60

8.3.6 Waste Sorting 61

9. Color Threshold Setting 63

10. Wireless Handle Control 67

10.1 Preparation 67

10.2 Device Connection 68

10.3 Control Mode Introduction 68

10.4 Button Function 69

10.4.1 Single Servo Mode 69

10.4.2 Coordinate Mode 70

11. Remote Desktop Installation and Connection 71

11.1 NoMachine Installation 71

11.2 Device Connection 74

11.2.2 Access via USB cable Using the Fixed IP 80

11.3 MobaXterm Installation 81

11.4 Connect Robot to MobaXterm 84

11.4.1 Connection 84

11.4.2 Open Multiple Dialogue Boxes 87

12. Position Adjustment for Object Gripping and Placing 89

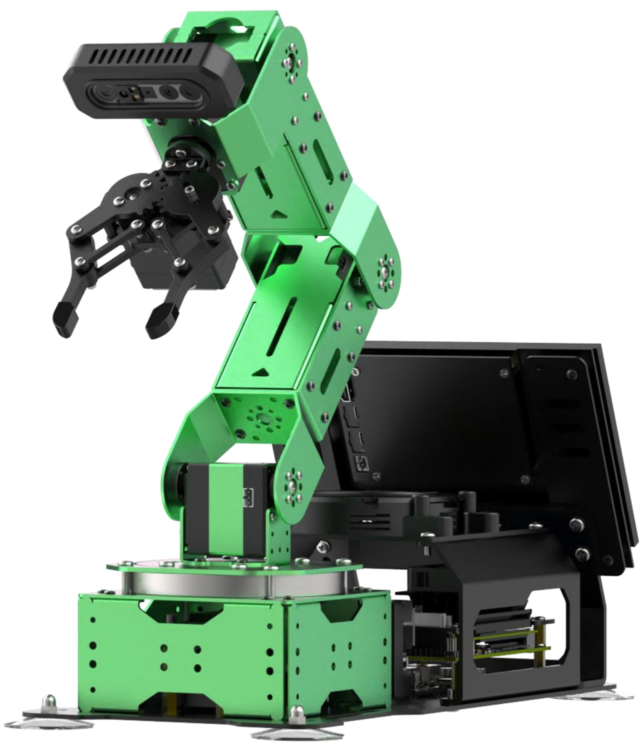
12.1 Preparation 89

1. Introduction

JetArm is an intelligent robotic arm featuring deep learning and computer vision capability.

It is outfitted with high-performance intelligent hardware, including NVIDIA Jetson Nano, 7-inch LCD display, 3D depth camera, and 6DOF robotic arm. With built-in inverse kinematics algorithms, it achieves high-precision AI recognition and precise grasping and placement functions across various scenarios, allowing it to perform sorting, stacking, classification, and more. Additionally, with microphone, speaker and sound card, JetArm ultimate kit extends its capabilities to enable voice-controlled color recognition and tracking.

In this chapter, JetArm ultimate kit with microphone and screen is used as an example for explanation, and the manual is equally applicable to other JetArm kits.



1.1 Robot Version Instruction

To satisfy different learning needs, we prepare 4 robot kits for your selection, respectively JetArm starter kit, JetArm standard kit, JetArm advanced kit and JetArm Ultimate kit.

1.1.1 JetArm Starter Kit

Starter kit: 6DOF robotic arm + USB monocular camera + control system + invisible antennas

It enables basic robot control, inverse kinematics and AI vision functionalities such as sorting and tracking for color and tag, machine learning, waste sorting.

1.1.2 Standard Kit

Standard kit: 6DOF robotic arm + Gemini depth camera + control system + invisible antennas

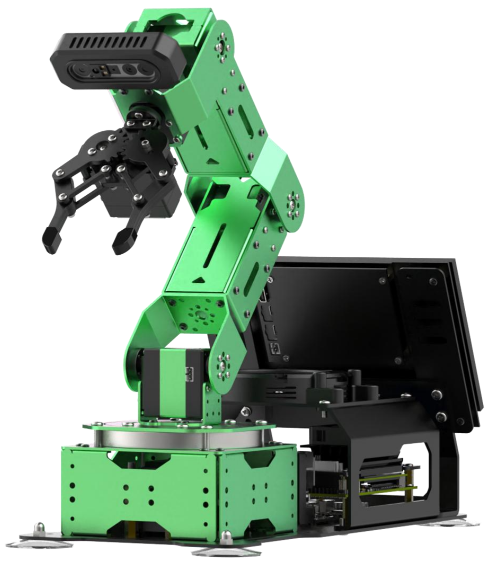
Except the functions of starter kit, it can perform distance measurement, 3D scene grabbing, sorting based on 3D shapes and other depth camera functionalities.

1.1.3 Advanced Kit

Advanced kit: 6DOF robotic kit + Gemini depth camera + control system + invisible antenna + 7 inch LCD screen

Based on standard kit, it is equipped with a screen, allowing you to conrol the robot controlled upon booting, which is equivalent to a mobile mini-computer.

1.1.4 Ultimate Kit



Ultimate kit:6DOF robot arm + Gemini depth camera + control system + invisible antenna + 7 inch LCD screen + microphone

On top of the features offered in advanced kit, ultimate kit includes a voice microphone kit that enables voice-controlled functionalities such as color recognition, tracking, and other interactive actions with the robotic arm.

1.2 Packing List

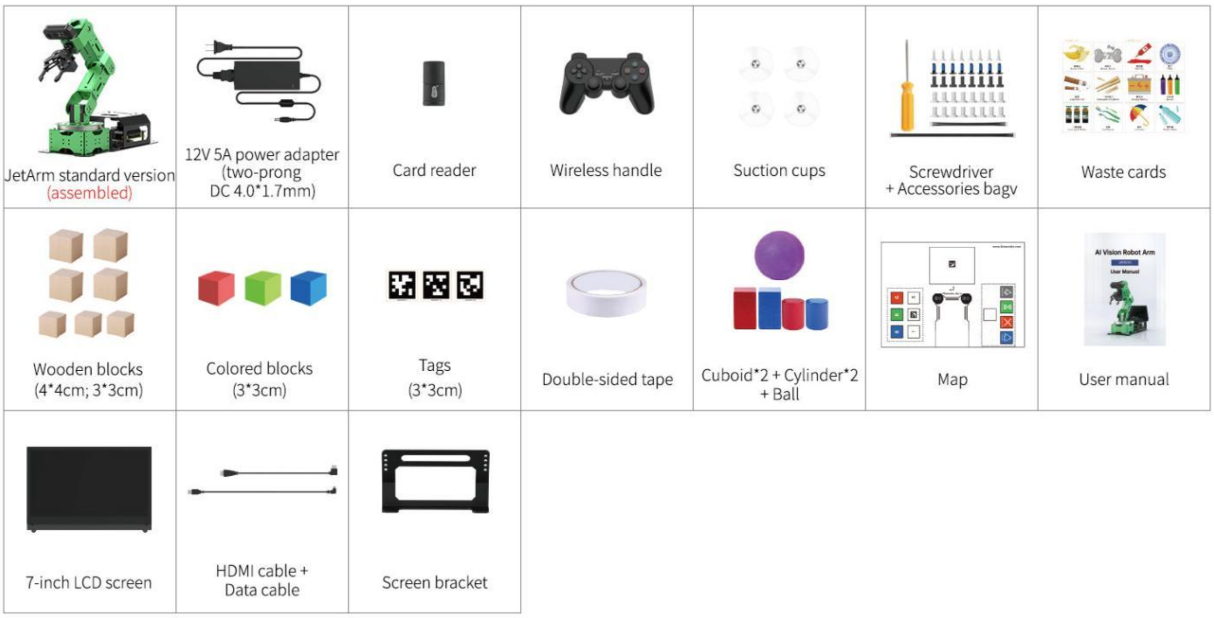
1.2.1 Starter Kit Packing List

Note: the handle receiver has already been plugged into USB hub before shipment.

1.2.2 Standard Kit Packing List

Note: the handle receiver will be plugged into the USB hub of robotic arm before shipment.

1.2.3 Advanced Kit Packing List



Note:

1. the handle receiver will be plugged into the USB hub of robotic arm before shipment.

2. the HD cable and date cable for screen will be connected to robotic arm.

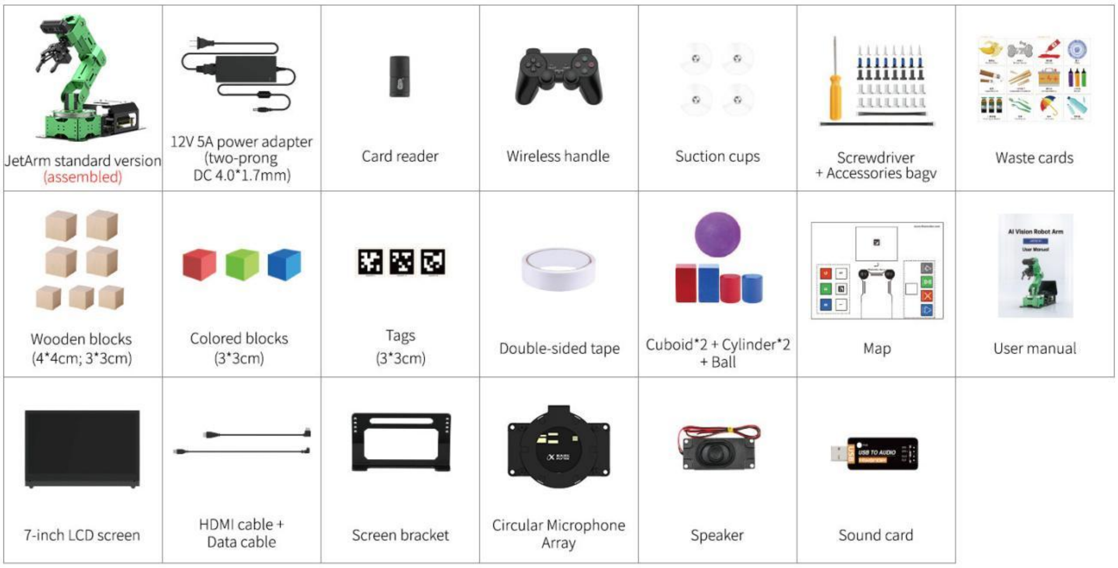
1.2.4 Advanced Kit Packing List

Note:

1.The handle receiver will be plugged into the USB hub of robotic arm during manufacturing.

2.The HD cable and data cable for screen will be connected to robotic arm.

3.The microphone, speaker and sound card will be installed on robotic arm.



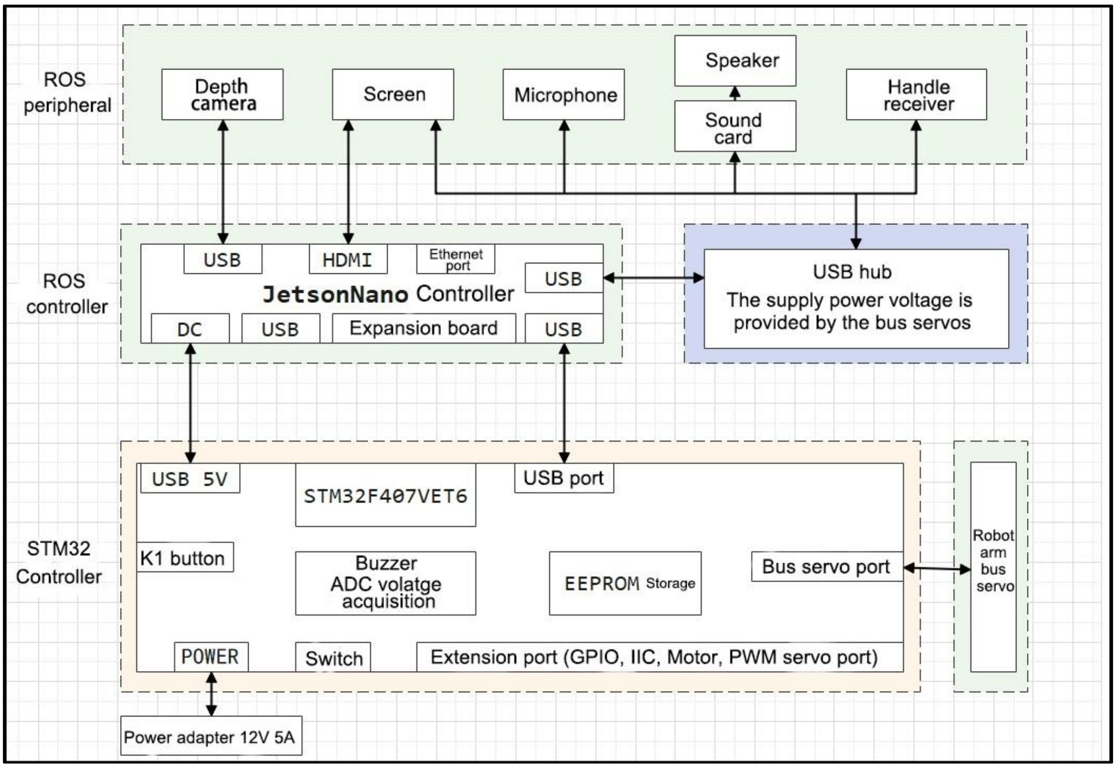
Note: Circular Microphone Array, speaker and sound card have already been installed on the robot arm before delivery.

2. Hardware Introduction

This chapter will mainly introduce the hardware components of ROS robots, including the electrical control system, ROS main control unit, depth camera and various sensors.

2.1 Hardware System

The system diagram for the hardware connection between the STM32-based robot and the ROS-based Jetson Nano robot is structured as follows.



The robot can be controlled using wireless handle, app or via ROS.

2.2 Electrical Control System Introduction

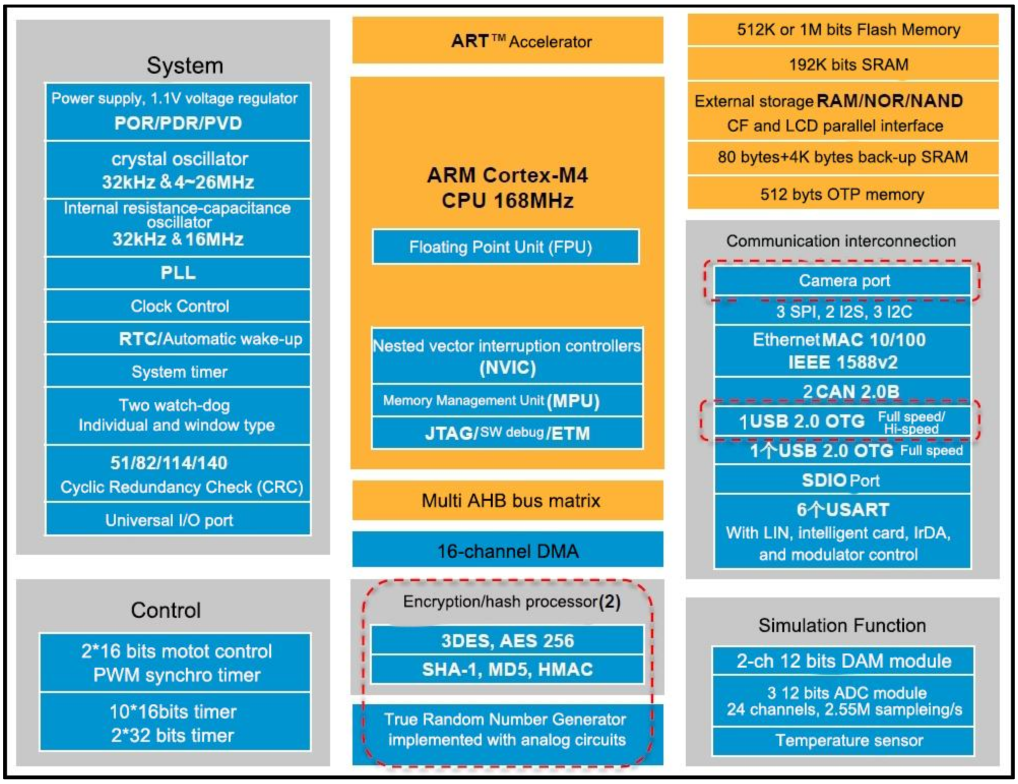
Utilizing an STM32 controller as the underlying motion controller, JetArm accomplish the communication between the Jetson Nano board and the bus servos.

2.2.1 STM32 Controller

The STM32 open-source controller is specially designed for robot development, small and exquisite.

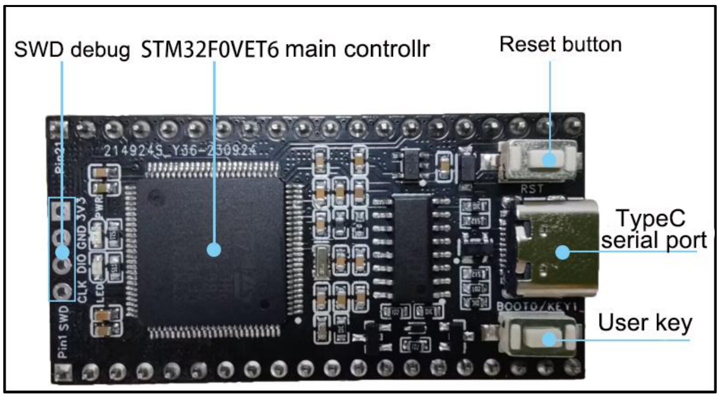
The controller deploys STM32F407VET6 as the main control, utilizing Arm's Cortex-M4 core, running at 168MHz. It has an onboard FLASH capacity of 512K, an SRAM capacity of 192K, and integrates FPU and DSP instructions.

The STM32F40x/41x series system block diagram is shown below:



With ample onboard resources and extension port, the controller is highly suitable for ROS robot development. It can be seamlessly integrated with the Jetson series ROS main control to make a ROS-based robot.

The front resource configuration of the control board is shown below：



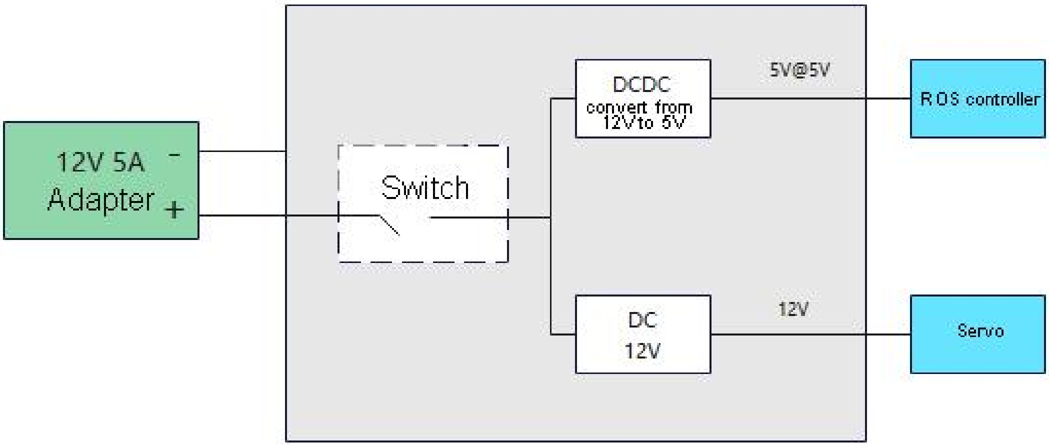
The control board also provides schematics and supports USB serial port program downloads, enabling STM32 programming for secondary development. It offers onboard resources and peripheral sample codes, facilitating user learning and application.

The resources and peripherals of STM32F407VET6 processor is shown as in the table below, and you can refers to the specific instruction in appendix.

2.2.2 Power Instruction

12V 5A adapter (DC4.0mm)

The robot system power is shown in the following figure. The 12V input from the battery is directly used to power the bus servos after passing through the STM32 controller. It is then converted to a 5V voltage through a DCDC converter to supply power to the Jetson Nano main controller, capable of delivering a maximum current of 5A, effectively meeting the power supply requirements of the Jetson Nano.

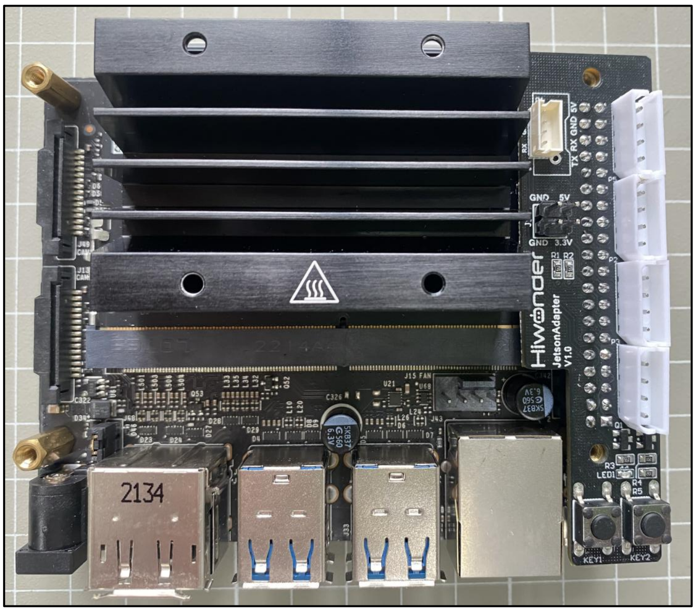


2.3 Hardware Module Instruction

In this case, JetArm ultimate kit is used an example to explain. Other kits with same hardware and function are applicable.

If you want to get detailed information for each module, please refer to "4. Hardware".

2.3.1 Main Controller



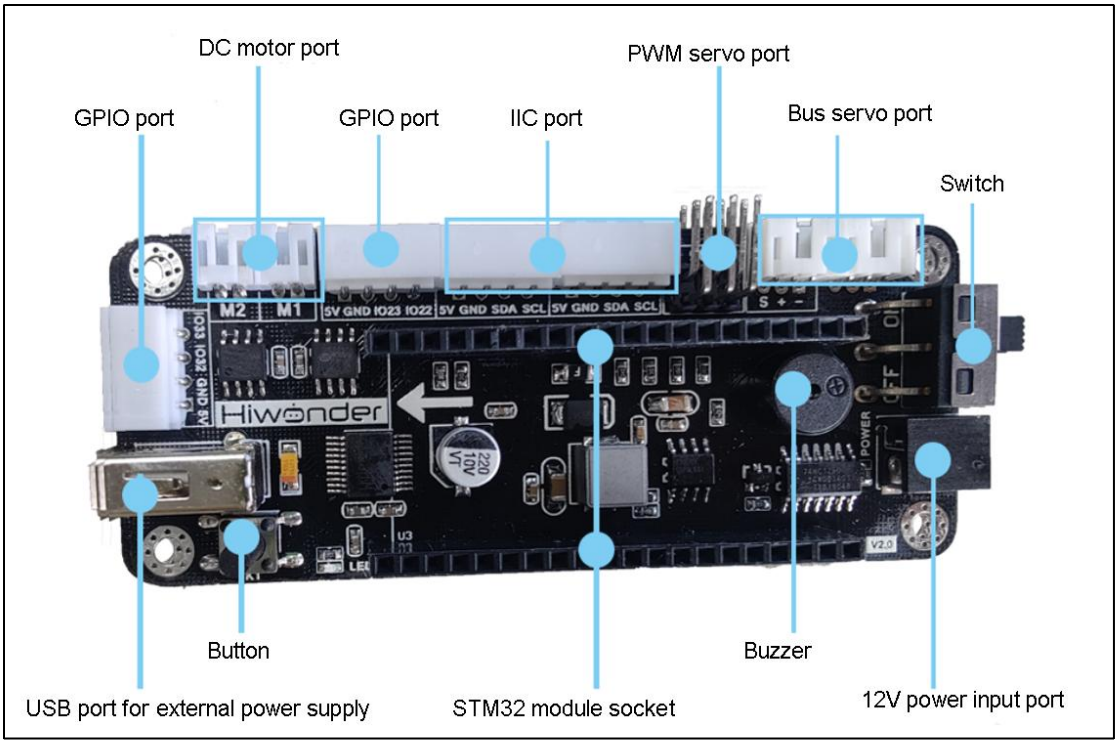
The robot employs the Jetson Nano as its ROS main control, composed of Jetson Nano board and Jetson Mini expansion board. The Jetson Nano board is a compact yet powerful computer capable of running mainstream deep learning frameworks, delivering the computational power required for most artificial intelligence projects.

The Jetson Nano board is installed the Ubuntu18.04 system and set up ROS Melodic environment. Regarding the basic Jetson Nano materials, please refer to “4. **Hardware Learning/ Jeson Nano Lesson**”.

The expansion board extends LEDs and buttons, facilitating users to check network status through LED blinking and switch network modes via mobile devices. Additionally, it reserves GPIO and I2C interfaces for further customization”.

2.3.2 Robot Driver Board

The front resource configuration of driver board is shown in figure below:



The driver board is responsible for supplying power to the robotic arm hardware, connecting to the bus servos, and interfacing with the STM32 control board to control servo movements.

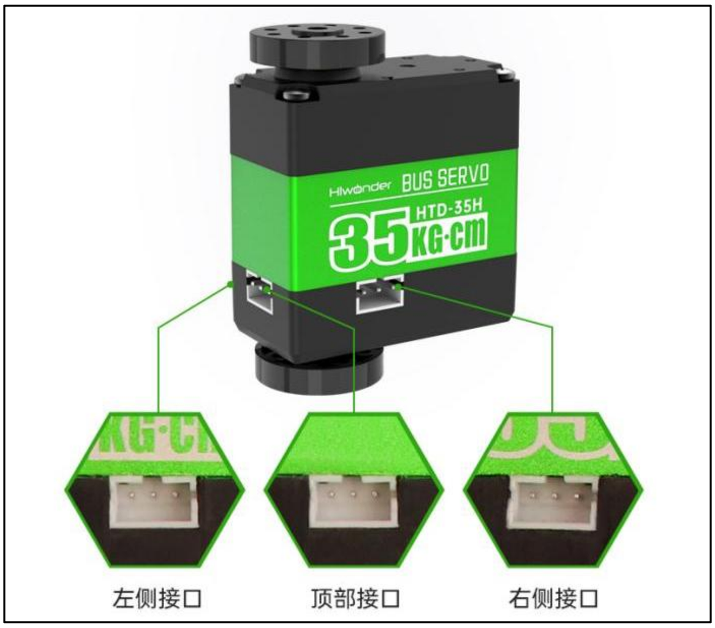
For further information of expansion board and STM32 controller, please refer to “4. Basic Control Lesson/ 1. STM32 Controller”.

2.3.3 Intelligent Bus Servo

JetArm is a 6DOF robotic arm composed of bus servos and interconnectd metal components.


The distribution of six bus servos: HTS-21H*1 (claw) + HX-12H (wrist) + HTD-35H*3 (body) + HTS-35H*1 (Pan-tilt)

Bus servos uses serial communication, connecting multiple servos to a control system via a single bus servo. They’re connected in series through an I/O port, allowing for higher precision compared to standard digital servos Here is an example of interface distribution and specifications using the HTD-35H servo model:



Pin	Instruction
GND	GND
VIN	VIN
SIG	signal terminal ("Half-duplex UART asynchronous serial interface)

2.3.4 Depth Camera

Camera serves as an essential component in robot's structure, which is equivalent to human's eyes.

JetArm employs the Orbbec Astra Gemini embedded 3D camera module, capable of capturing object depth and color images. The compact size and high-performance advantages are suitable for 3D object scanning terminals with a recognition distance of 25 to 250 centimeters. It supports a USB3.0 interface. With an active dual-camera structured light design, the Gemini depth camera provides high-resolution depth and infrared image information while demonstrating strong resistance to light interference.

In robotics, it is mainly applied for AI vision functions, deep learning, KCF object tracking. For the basic lesson of depth camera, you can find in **"6. Camera Basic Lesson/ 1. Depth Camera Basic Lesson**". For depth camera application, you can find in "**7. ROS+OpenCV Lesson, 8. ROS+Deep Learning Basic Lesson, ROS+Deep Learning Application, 11. Robotic**

Arm Depth Camera Application, 12. Voice Control Lesson/2. Voice Control".

The specific camera parameters are shown in the following table:

<table>
<tr><th>Name</th><th>Parameter</th><th colspan="2">Instruction</th></tr>
<tr><td rowspan="7">Depth parameters</td><td>Depth technology</td><td colspan="2">ORBBEC® Binocular structure light</td></tr>
<tr><td>Working range</td><td colspan="2">0.25~2.5m</td></tr>
<tr><td>Accuracy</td><td colspan="2">1m：±5mm</td></tr>
<tr><td>Field of view (FOV)</td><td colspan="2">H 67.9°×V 45.3°×D 78°±3.0°</td></tr>
<tr><td>Resolution @ Frame Rate</td><td>USB2.0 mode：
1280×800@7fps
640×400@30fps</td><td>USB3.0 mode：
1280×800@30fps
640×400@60fps</td></tr>
<tr><td>Depth processing chip</td><td colspan="2">MX6000</td></tr>
<tr><td>Close-range protection</td><td colspan="2">Not support</td></tr>
<tr><td>RGB</td><td>Field of View (FOV)</td><td colspan="2">H 71.50°×V 56.7°×D 84.00°</td></tr>
</table>

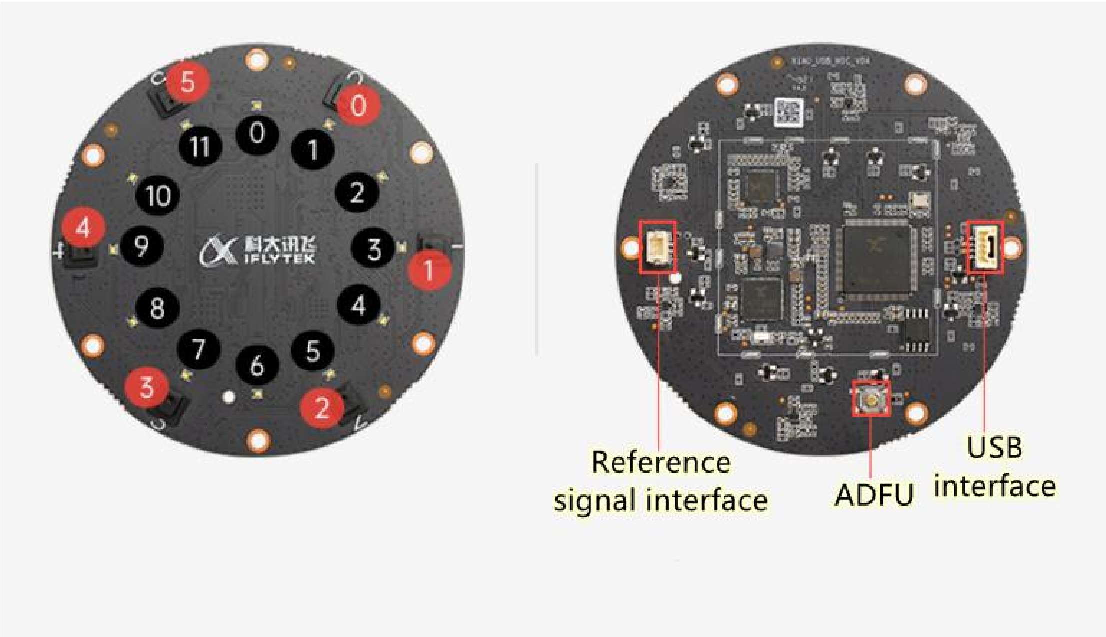
	Resolution @ Frame Rate	USB2.0 Mode： 1280×720@7fps 640×480@30fps	USB3.0 Mode： 1920×1080@30fps 1280×720@30fps 640×480@30fps 640×480@60fps
	UVC	Support	
Others	Baseline	40mm	
	Operation System	Support Android/Linux/Windows	
	Working Environment	Indoor / semi-outdoor (specific to application scenarios and related algorithm requirements)	
	Data port	USB3.0 Type-C	
	Power supply	USB3.0 Type-C	
	Microphone	/	
	Power consumption	Average power consumption: 2.2W Standby power consumption: 0.9W Peak Power consumption: 5W	
	Working temperature	10℃~40℃	

	Security	Class1 laser
	Dimension	65.30*22.5*12.30

2.3.5 Microphone Module

The microphone module is optional in robot kits. With 6-microphone array, JetArm can realize voice wake-up and voice control functions. For more information of microphone, please refer to "**12. Voice Control Lesson**".

The 6-microphone array is an advanced audio system that consists of six microphones, allowing for 360° recording capabilities. This array can sample and process spatial features of a sound field, enabling it to locate sound sources, suppress background noise, interference, reverberation, and echo, as well as extract and separate signals with precision.



Interface	Function
USB interface	Connect to PC or embedded devices

ADFU	Used for Flash. To enter developer mode, connect the array and computer through USB cable while pressing this key.
Reference signal interface	cancel echo
Microphone	Numbered from 0 to 5
LED	Numbered from 0 to 11 clockwise

2.3.6 Sound Card and Speaker

JetArm Ultimate kit comes with sound card and speaker. They're collaborated with 6-microphone array, enabling the functionality of voice interaction with JetArm.

The parameter instruction for sound card and speaker is shown in the list below:

Sound card parameters instruction:

Supply Power Voltage	5V

Audio codec chip	SSS1629A5
Control interface	USB
Audio interface	PH2.0
Speaker driver	2.6W per channel (4Ω BTL)

Speaker parameters instruction:

Impedance	4Ω
Peak Power	2W

2.3.7 Screen

(The system desktop is for your reference only. Please subject to the actual situation)

JetArm Ultimate kit comes with 7-inch touch screen with both display and touch capabilities. Upon booting up, you can directly access the robot's system desktop, and quickly experience PC software by clicking its icon. With keyboard and mouse, robot is controlled directly without the need for a remote connection.

Regarding the related PC software lesson, please refer to “4. **Robotic Arm Control**”.

Display dimension	7 inch
Resolution	1024*600px
Interface	USB/HDMI
NAVIDA/Raspberry Pi/Ubuntu Image	Support

3. Accessory Assembly

Upon receiving JetArm ultimate kit, you need to install 7-inch LCD screen and suction cups on robotic arm.

3.1 Install 7-Inch LCD Screen

Note：Assembly instructions take JetArm ultimate kit as example. They are also applicable to other robot kits.

Please prepare 7-inch LCD screen, screen bracket and screws.

Note:

1) Attach the 7-inch LCD screen to the screen bracket with M4*6 machine screws.

Installation effect

2) Fix the screen bracket on the bottom with M4*6 screws.

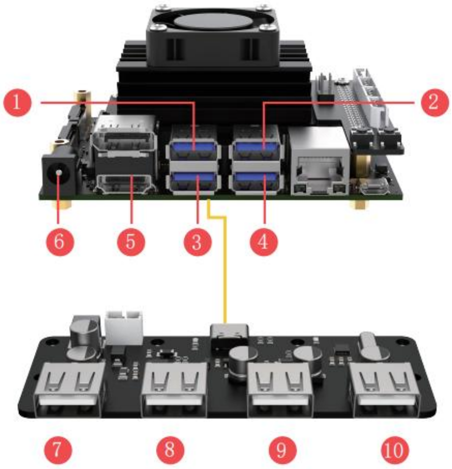
Installation effect

3) Connect HDMI cable to HDMI interface on the 7-inch screen, and connect USB cable to CTOUCH port.

Note: 7-inch LCD screen does not support touch function if the cable is not connected to CTOUCH port.

Installation effect

3.2 USB Interface Instruction



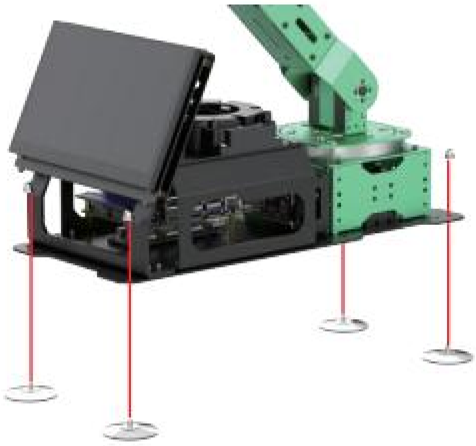
No.	External Module
1	Custom expansion development
2	STM32 controller commnication cable
3	USB hub communcation cable
4	Depth/monocular camera
5	7 inch screen HDMI cable (advanced kit and ultimate kit)
6	STM32 controller power cable

Module Connected to USB Hub as follow:
(If the modules have been connected, you just need to confirm)

No.	External Module
7	Sound card+speaker (ultimate kit only)
8	6-ch Microphone array module (ultimte kit only)
9	7-inch screen power cable (advanced kit + ultimate kit)
10	Handle receiver

3.3 Install Suction Cups

Prepare suction cups and nuts. Insert the suction cups from below and tighten the nuts.



Installation effect

4. Initial Startup and Test

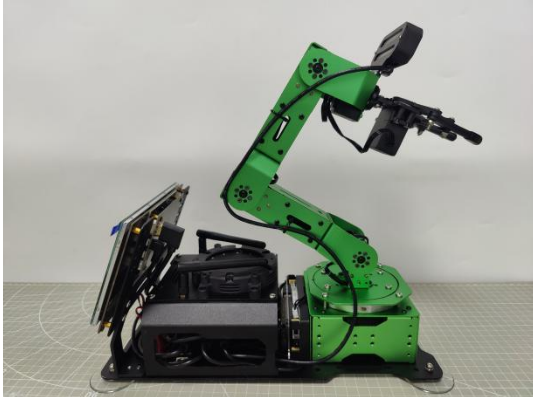
4.1 Note

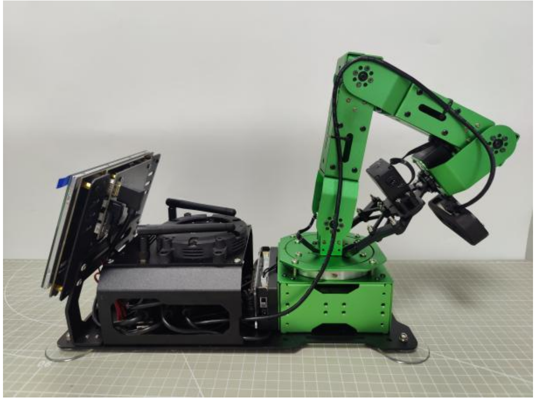
1) Place robot on a flat and smooth surface.

2) Ensure that the cables connected and handle receiver are properly connected.

3) Keep a safe distance from the robotic arm before powering it on to avoid injury.

4) Robotic arm will present initial pose after staring. To prevent damage caused by sudden servo force, please avoid stacking the servos on the top of each other before turning on the robot.





5) After powering on, please do not forcefully manipulate the servos to prevent any damage to them.

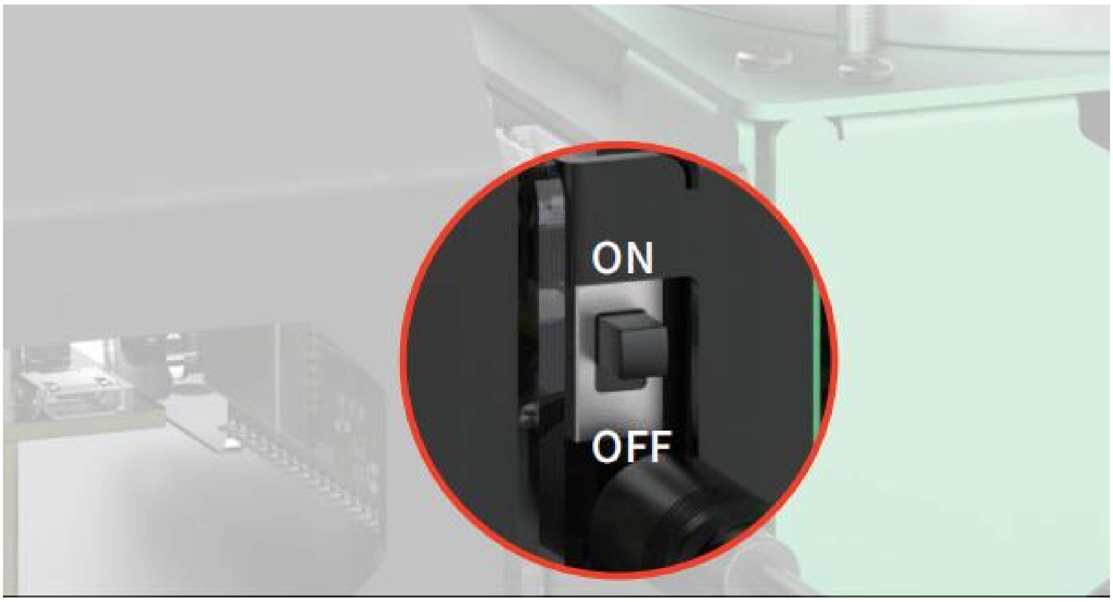
4.2 Connect Power Adapter

Prepare 12V 5A power adapter, then connect it to the power port on the driver board, as shown in the figure below:

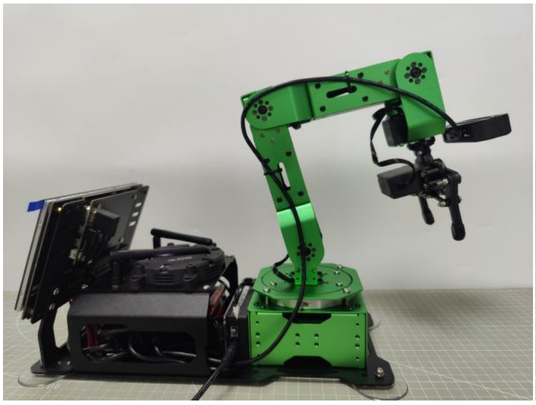
4.3 Robot Boot-up Status and Testing Instruction

4.3.1 Boot-up Status

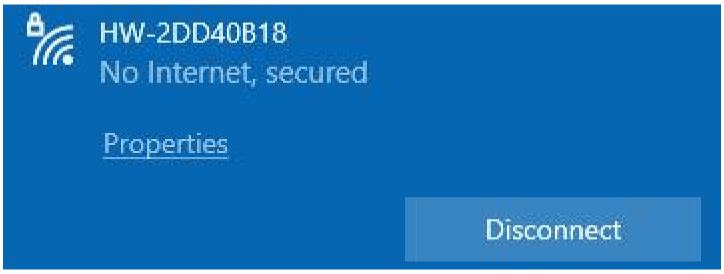
1) Turn on the switch on driver board (push it upwards). At this point, LED on Jetson Nano mini control board will light up blue. Once the network configuration is completed, the blue LED will flash every 1 second. The startup process takes approximately 35 seconds, please be patient.

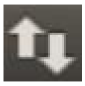


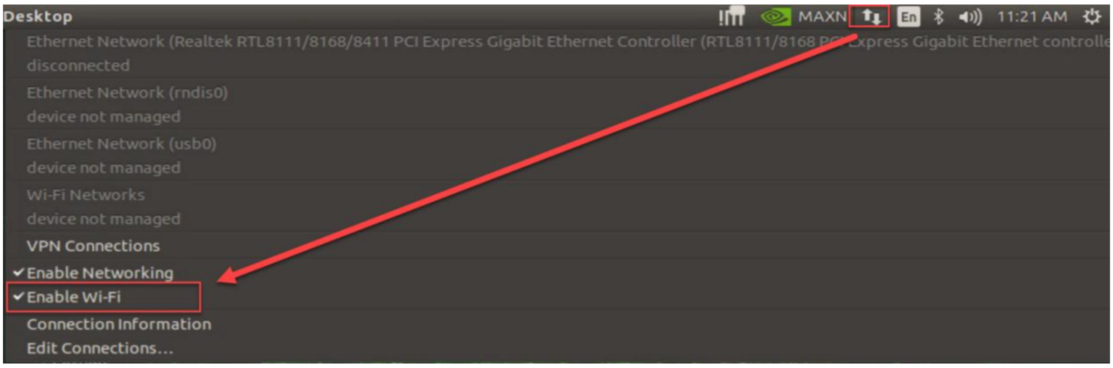
2) When robot arm restores its initial stance, it generates a beep soun, the speaker broadcasts “I’m ready”, and the screen displays system desktop, which indicates a successful startup of robot.



3) Robot is in AP direct connection mode by default. After the robot is powered up, it will generate a WIFI starting with “HW” with an initial password of “hiwonder” for app connection and remote desktop system connection.



Warming Tip: if robot’s WIFI cannot be found, please check if the antenna on Jetson Nano has been disconnected. Alternatively, click  **on the top right of the screen to check if “Enable Wi-Fi” is selected （if not selected, please manually select it）.**



4) If you’re using JetArm ultimate kit, you can hear the broadcast of “I’m ready”.

4.3.2 Hardware Test

You can refer to the following table to test hardware:

Module	Test Operation	Outcome
Bus servo	After connecting	Upon turning on the

	servos, turn on robot.	robot, it presents an initial stance.
7-inch touch screen	Ensure the operations below: 1) The screen HDMI cable is properly connected. 2) The power cable is connected to the C-TOUCH interface. Test: tap the icons in system desktop.	Normal display of desktop and touchscreen functionality
Sound card+speaker	Ensure the USB cable of sound card and speaker are connected properly.	After robot starts up, it will broadcast "I'm ready".
6-microphone array module	Ensure the USB cable of microphone is connected properly.	Light up a LED after powering up.
Depth camera	1) Open WonderAi and connect with robot. 2) Select "Robot Control", then the app interface will display the live camera feed.	Display the live camera feed.

5. Installation and Connection

Users can control robot using app “WonderAi”. Following will specify the operations.

Note:

1) Please ensure all app permissions are enables, otherwise app functions will be limited.

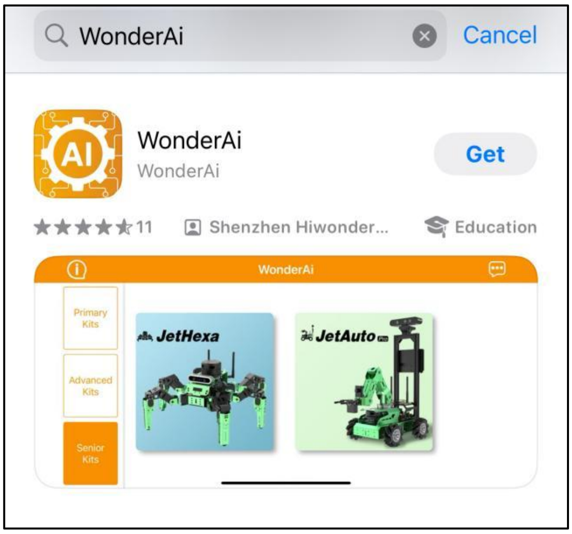
2) Before opening app, it is necessary to enable GPS and WIFI functionalities on phone settings.

5.1 App Installation

5.1.1 APP Store Download

1) Android system: the app installation package is located in “**2. Software Tool/ 1. App Installation Package**”.

2) iOS system: search and install “**WonderAi**” in App Store.



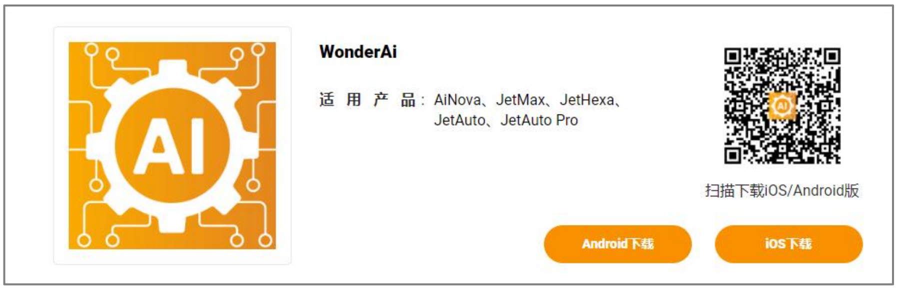
5.1.2 Hiwonder Official Website

1) Enter Hiwonder official website “https://www.hiwonder.com.cn/” in

browser.

2) Click “**Software Download**”, then click “**APP**”.

3) Find “**WonderAi**”, then you can scan the corresponding QR code to download the app.



5.2 Connection Mode Introduction

There are two network modes, including AP direct connection mode and STA LAN mode.

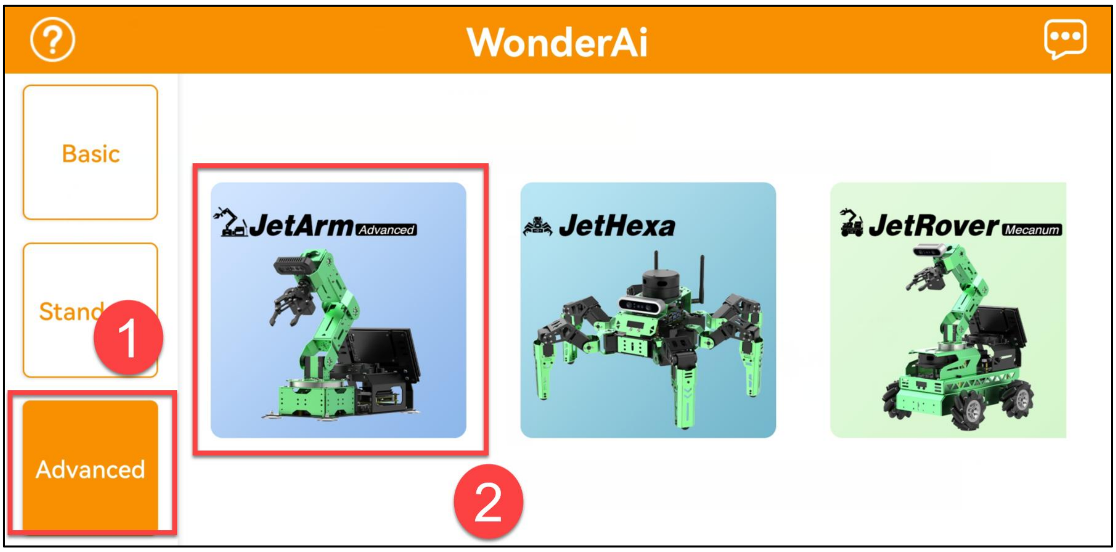
① AP direct connection mode: Jetson Nano generates a WiFi which can be connected by phones. This WiFi has no internet access.

② STA LAN mode: Jetson Nano actively connects to specific WiFi. You can access Internet in this mode.

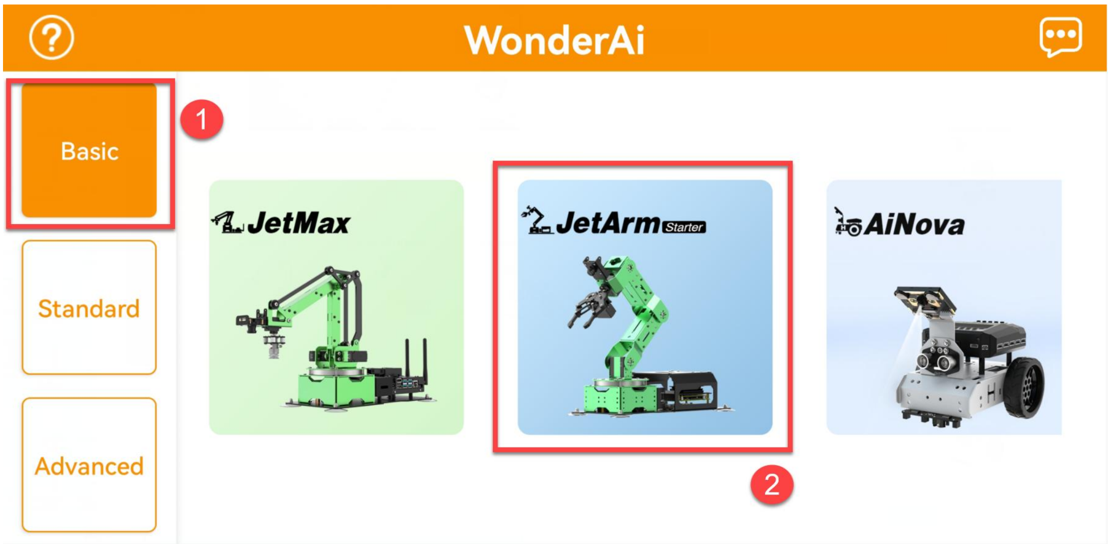
5.2.1 Direct Connection (Must Read)

Take JetArm advanced kit as example using Android system for demonstration. The operations are applicable to the iOS system and other robot kits.

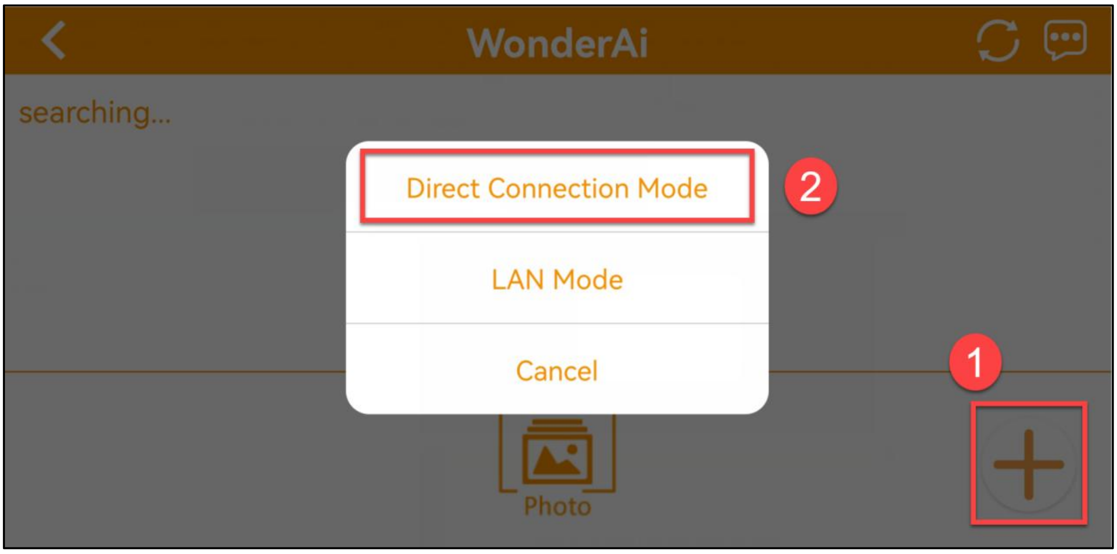
1) Open "**WonderAi**" APP, then tap "**Advanced->JetArm**" in sequence.



Note: if you're using JetArm starter kit, please tap "**Stater->JetArm**" in sequence.

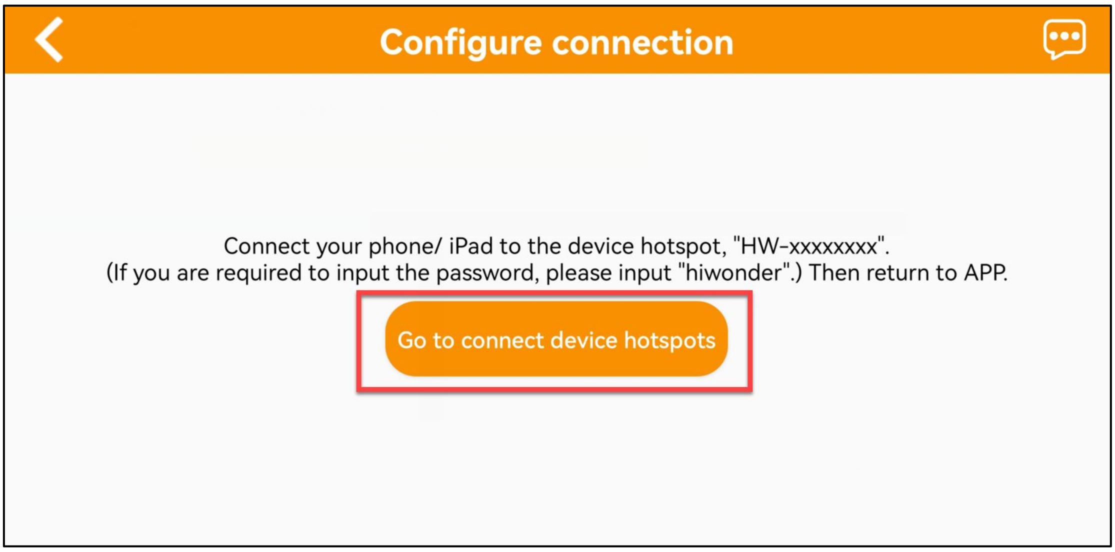


2) Click "**+**" button, and select "**Direct Connection Mode**".

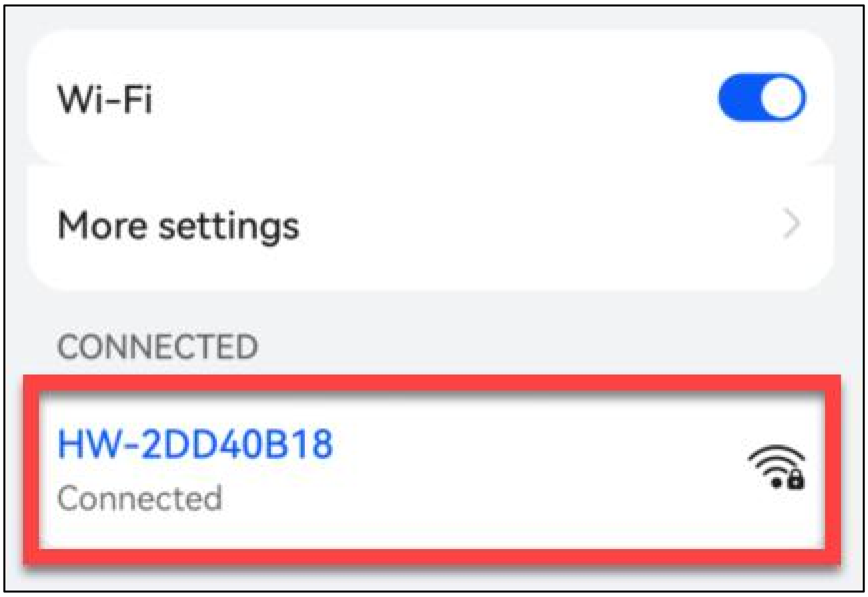


Note: for LAN mode connection, please refer to "5.2.2 LAN Mode Connection (Optional)".

3) Next, click "**Go to connect device hotspots**" to join the WiFi generated by JetArm.

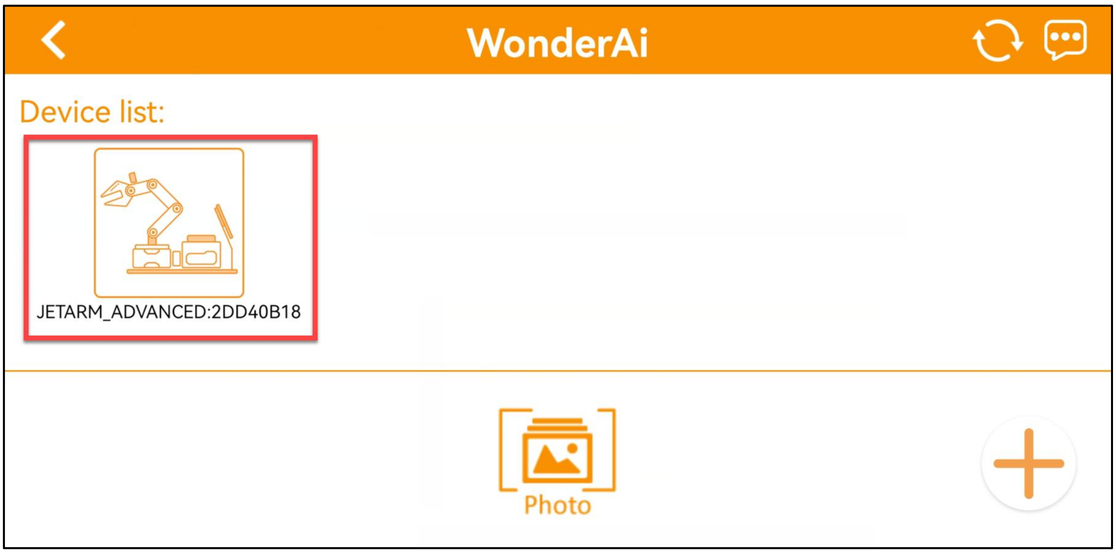


4) The device WiFi starts with “**HW**” and the password is “**hiwonder**”.



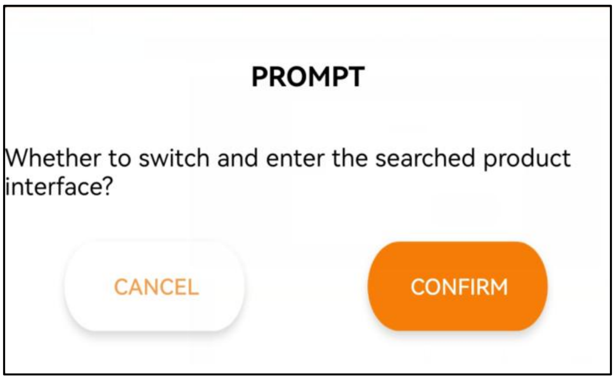
Note: for iOS user, please do not return back to the APP until the WiFi icon  **appears above, otherwise JetArm cannot be searched. If JetArm cannot be searched, please click**  **to refresh.**

5) Return to the APP, and click the robotic icon to enter home interface.

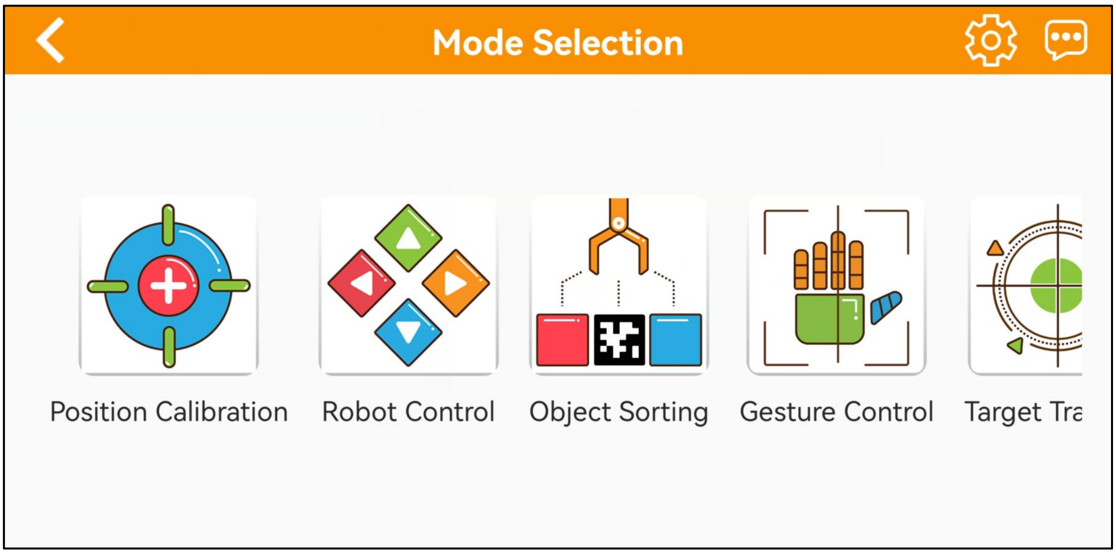


If you are informed of "No Internet Connection. Whether to keep connection", just select "Keep Connection".

6) When the following window pops up, it means that you have selected the wrong version. Click "**Confirm**", then the APP will automatically switch to the home interface of the right version.



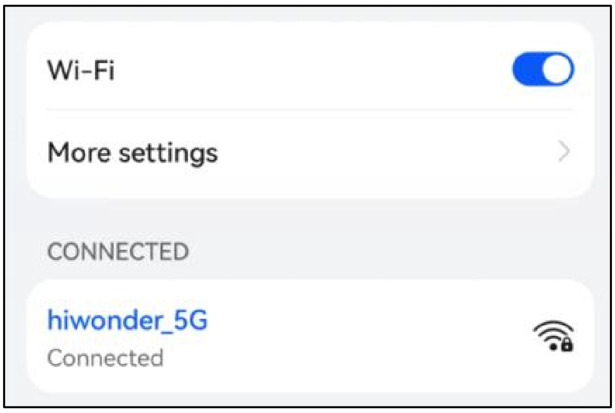
7) The home interface of the ultimate kit is as pictured.



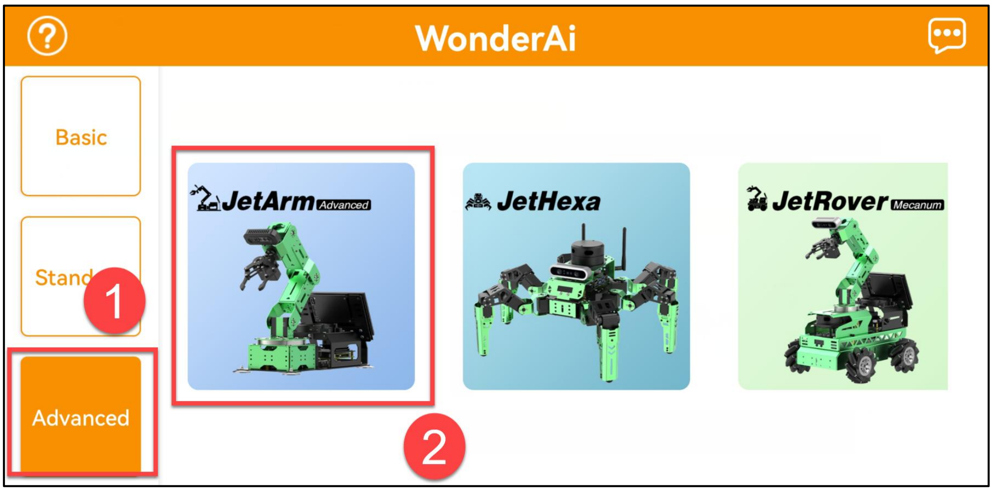
For the detailed operation method of each mode, please refer to “**8. APP Control**”.

5.2.2 LAN Mode（Optional）

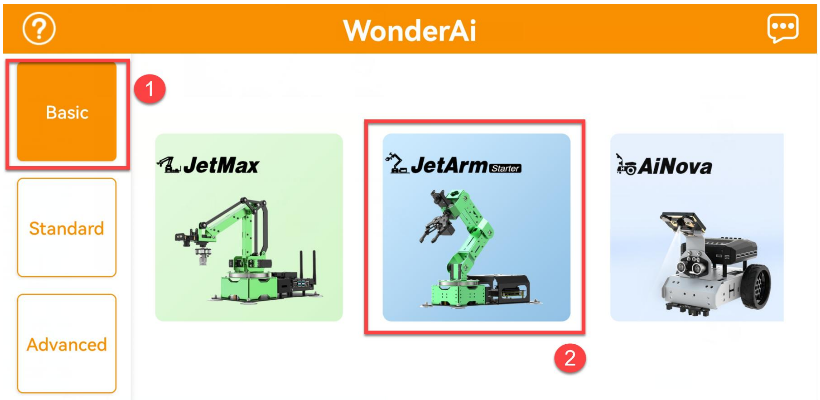
1) Firstly, join 5G network, for example “Hiwonder_5G”. (The router supporting dual-frequency will distinguish the Wi-Fi name by default under the situation that 2.4 G and 5G are separated. For example, Wi-Fi “Hiwonder” is 2.4 frequency band while “Hiwonder_5G” is 5G frequency band)



2) After connection, open WonderAi. Then click “**Advanced->JetArm**” in sequence. (Take JetArm advanced kit for example)

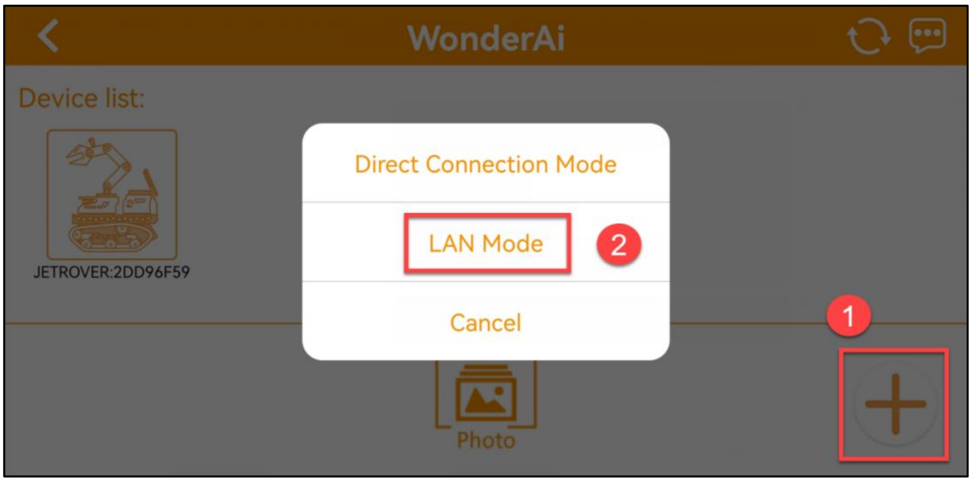


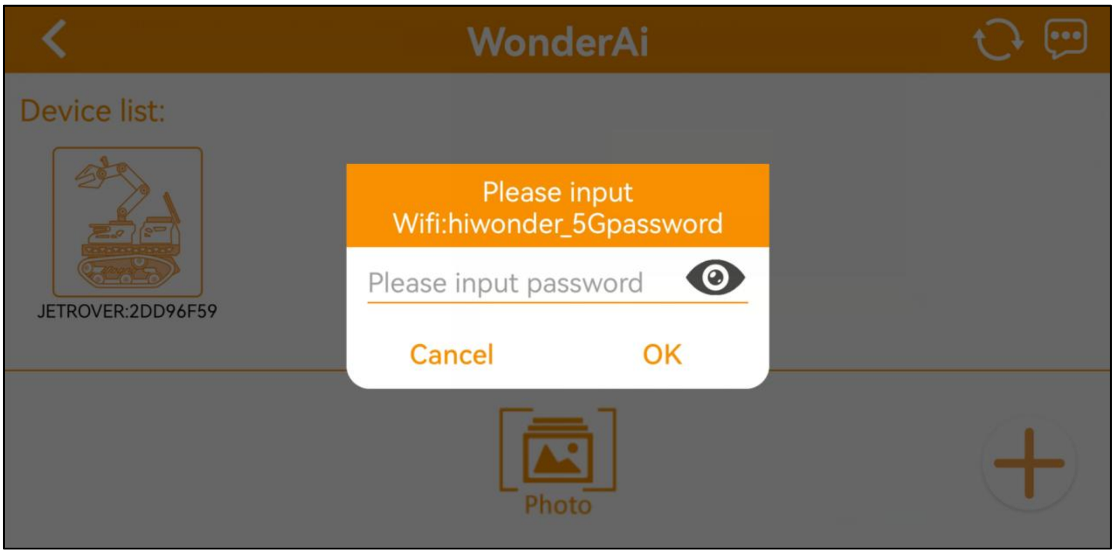
Note: if you're using JetArm starter kit, please tap "**Stater->JetArm**" in sequence.



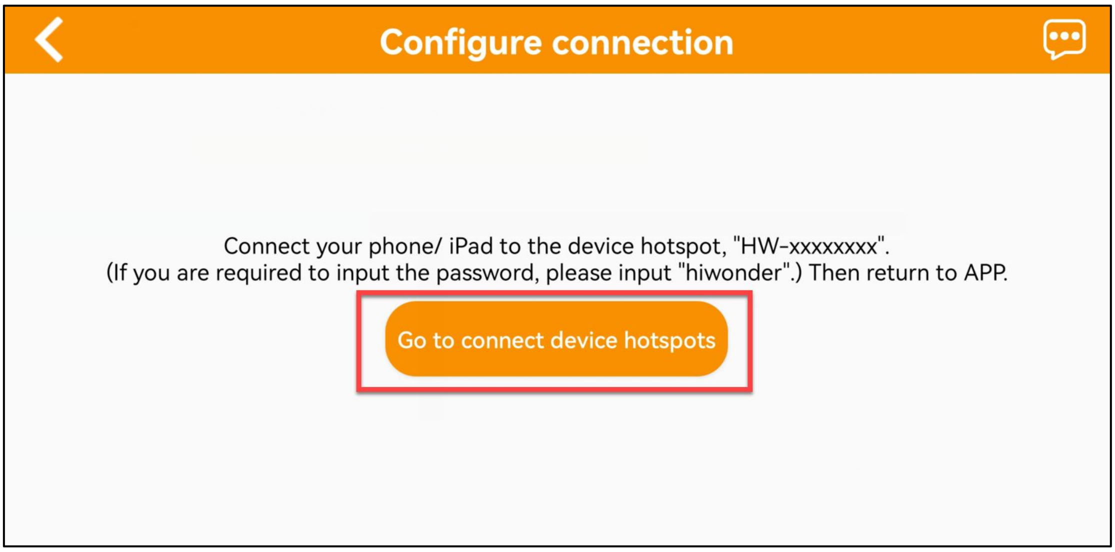
3) Continue, enter the Wi-Fi password. Having entered the password, click "**OK**". Please ensure the password you enter is correct.

4) Click "+" button at the lower right corner, then select LAN Mode.



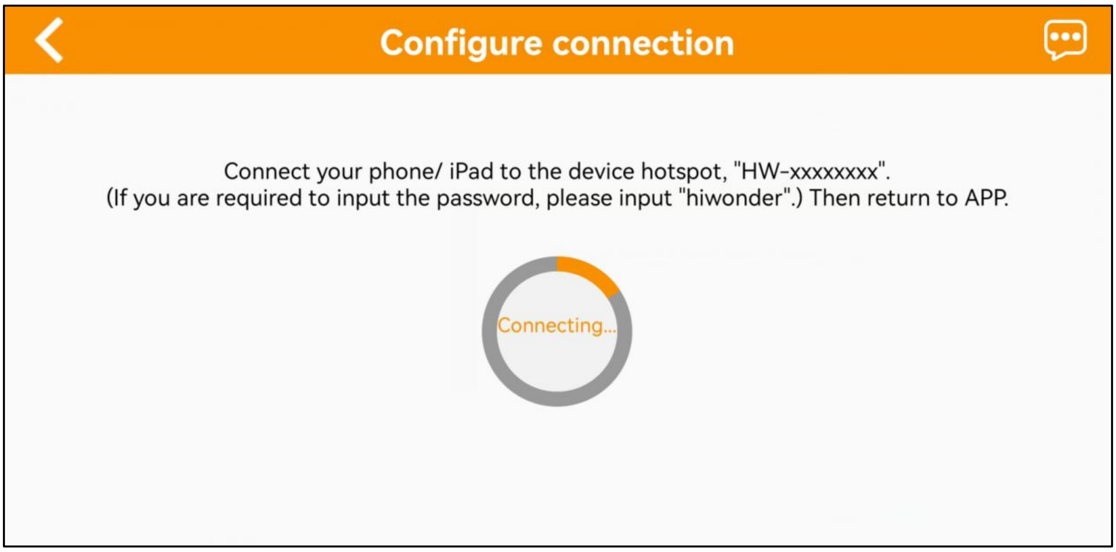


5) Click “**Go to connect device hotspots**”.

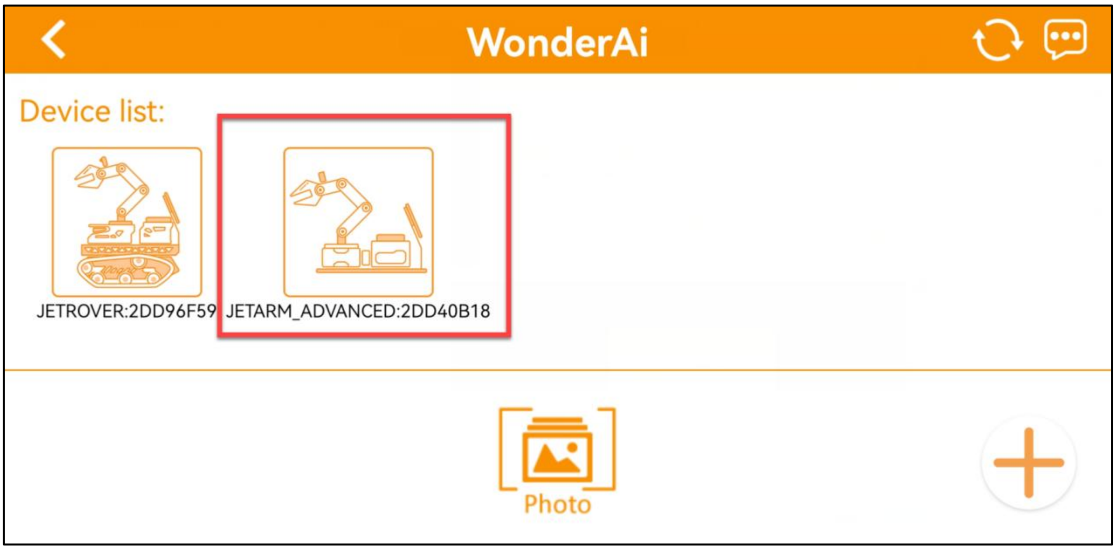


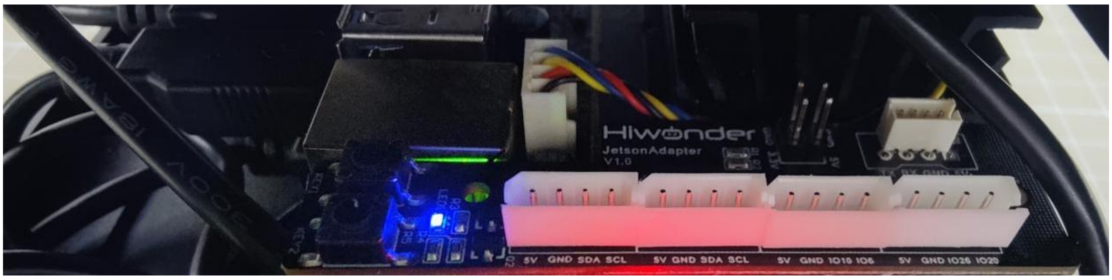
6) Join the WiFi starting with HW, and input the password “**hiwonder**”. After connection, return back to the APP interface.

7) Then, WonderAi APP is connecting to the robot.

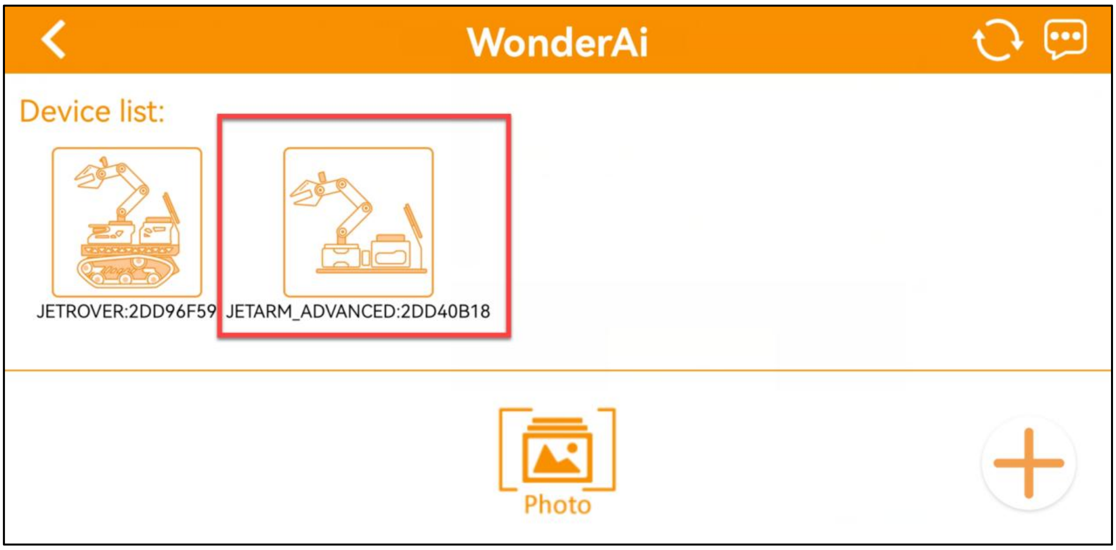


8) After a while, the robotic icon will show up. And at the same time, LED1 on the Jetson Nano mini expansion board will light up.



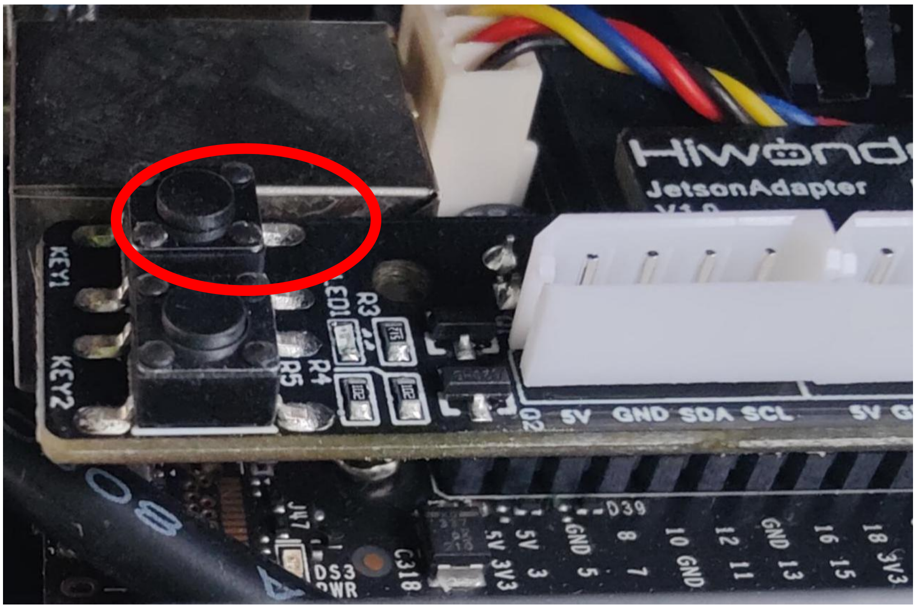


9) By long pressing the robotic icon, you can check the IP address and ID.



10) Input this IP address in the search bar on remote desktop software. Then you can enter the robot system desktop. For the operation method, please refer to the following content "**11. Remote Connection Tool Installation and Connection**".

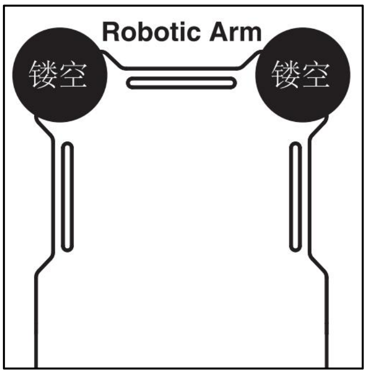
11) If you want switch back to Direct Connect Mode, press KEY1 button on Jetson Nano mini expansion board until blue LED flashes, which means the current mode is Direct Connect Mode.

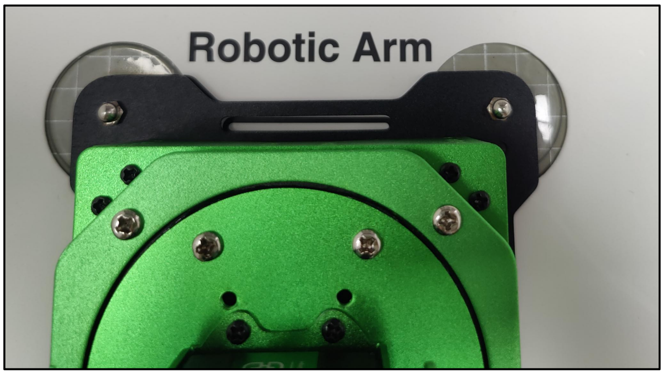


6. Map Introduction and Device Placement

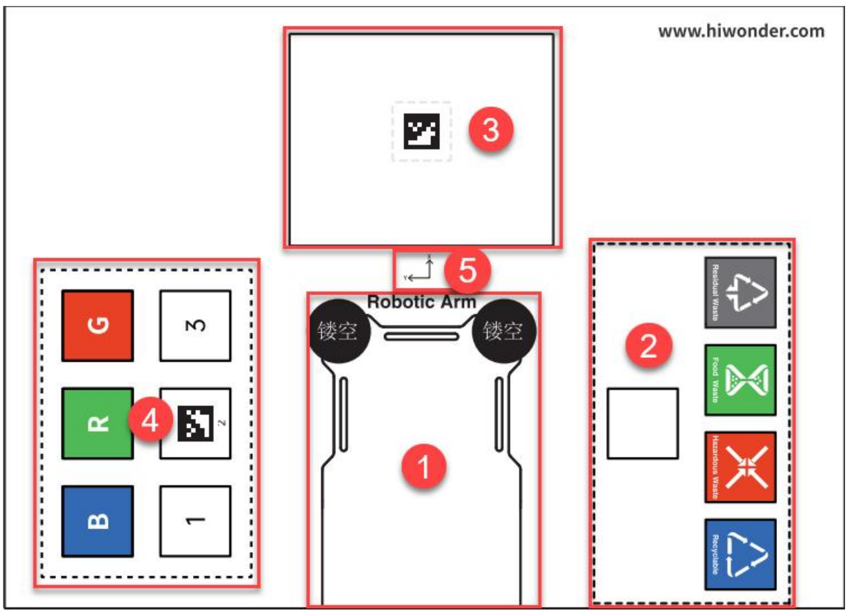
For a better experience, we need to have a basic understanding of the map. This section will introduce the placement of device and the distribution of map.

Firstly, lay the map flat on the table, and then position the robotic arm within the area as shown in the left illustration below. The front end of the base should align with the front boundary of the robotic arm placement area, and the suction cups should be firmly against the table, as depicted in the illustration on the right below.

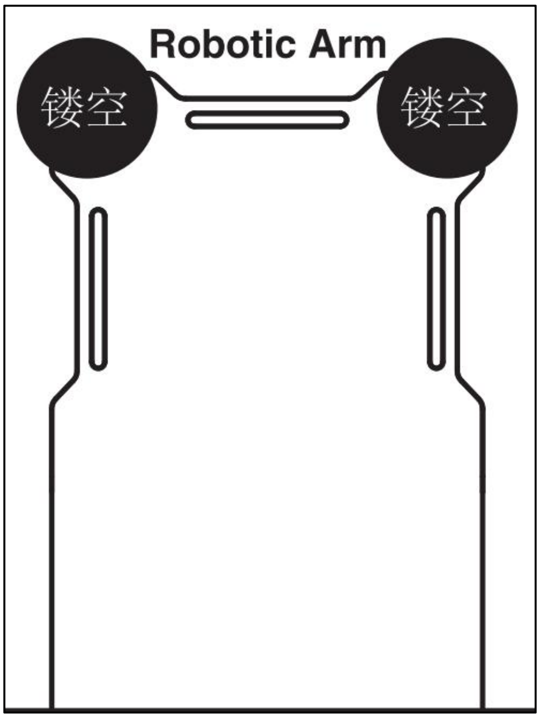




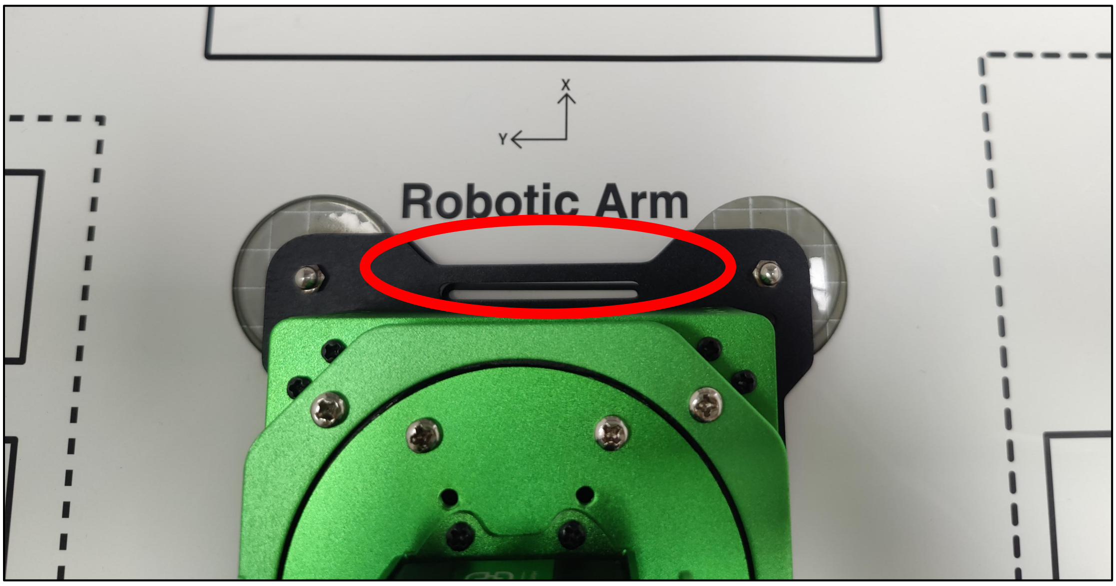
The map is composed of four sections: ① Robotic arm placement area ② Waste sorting area ③ Vision recognition area ④ Color and tag sorting area ⑤ X-axis and Y-axis coordinates of the robotic arm



① Robotic arm placement area

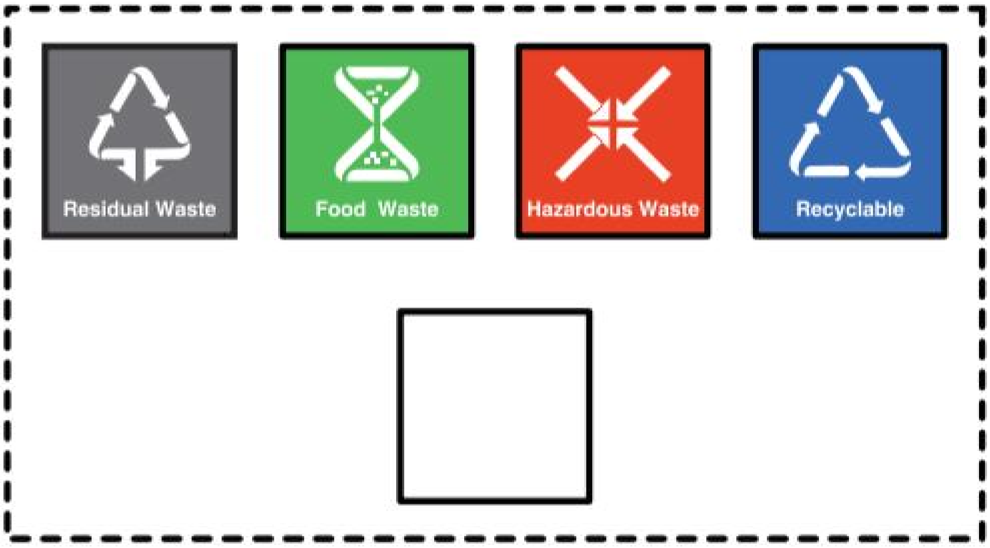


When placing the robotic arm, the front end of its base should align with the front boundary of the robotic arm placement area, and the suction cups should be firmly against the table, as depicted in the picture below.

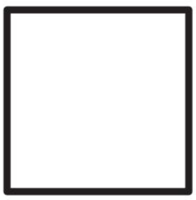


② Waste sorting area

The robotic arm will transport the blocks or cards from the recognition area to their respective placement area.



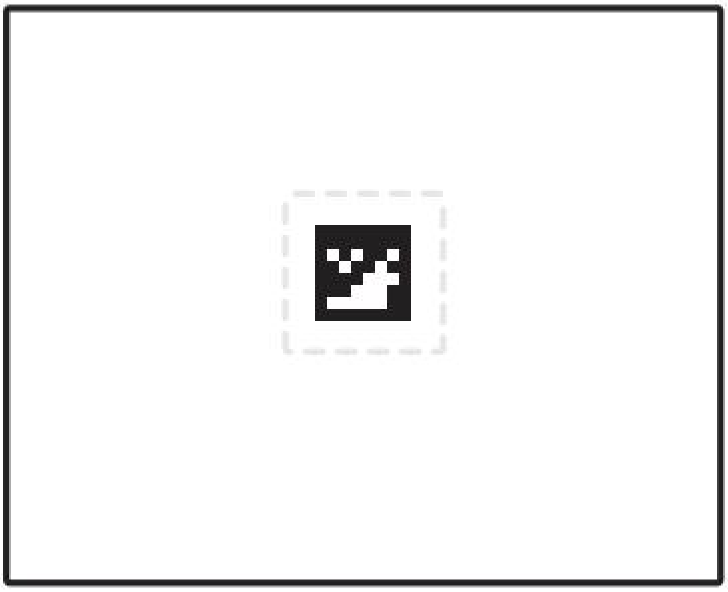
The functions of above areas are as follow.

Icon	Function
	Stacking area
	Residual waste cards placing area
	Food waste cards placing area

	Hazardous waste cards placing area
	Recyclable waste cards placing area

③ Vision Recognition Area

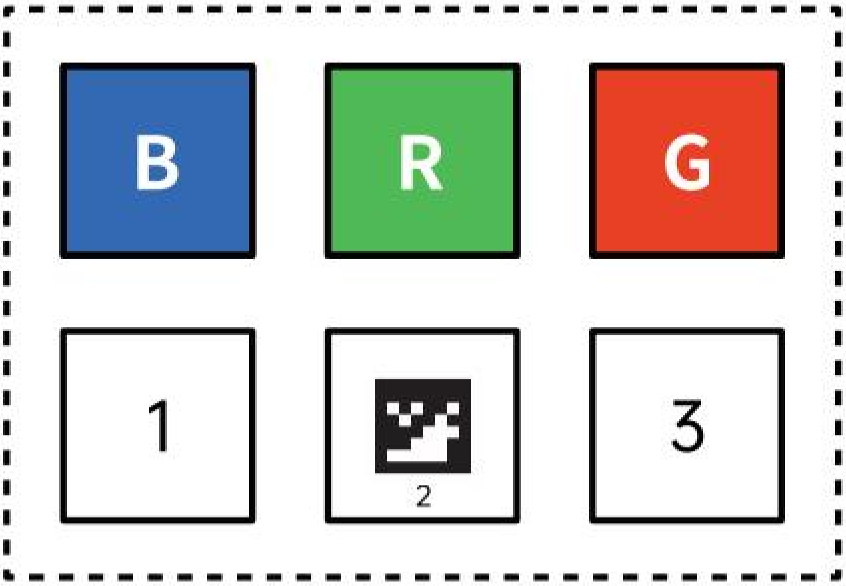
In robotic arm vision recognition area, the central label servers as a reference point during the position calibration of robot arm (for details on position calibration, please refer to "7. Position Calibration"). Place the blocks with garbage card or color blocks to be recognized within this area.



Note: when placing the blocks, do not locate them at the edges of the vision recognition area to prevent recognition failure .

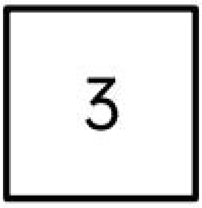
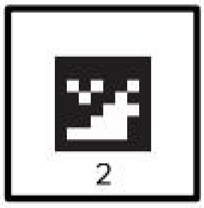
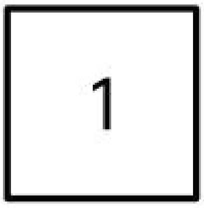
④ Color and tag sorting area

The robotic arm will transport the blocks from the recognition area to their respective placement area.



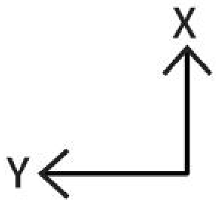
The functions of above areas are as follow.

Icon	Function
	Red block placing area
	Green block placing area
	Blue block placing area

	Tag ID3 block placing area
	Stacking and tag ID2 block placing area
	Tag ID1 block placing area

⑤ X-axis and Y-axis coordinates of robotic arm

After the robotic arm is placed on the map, you can refer to this coordinate axis to confirm the positive directions of X-axis and Y-axis of the robotic arm.

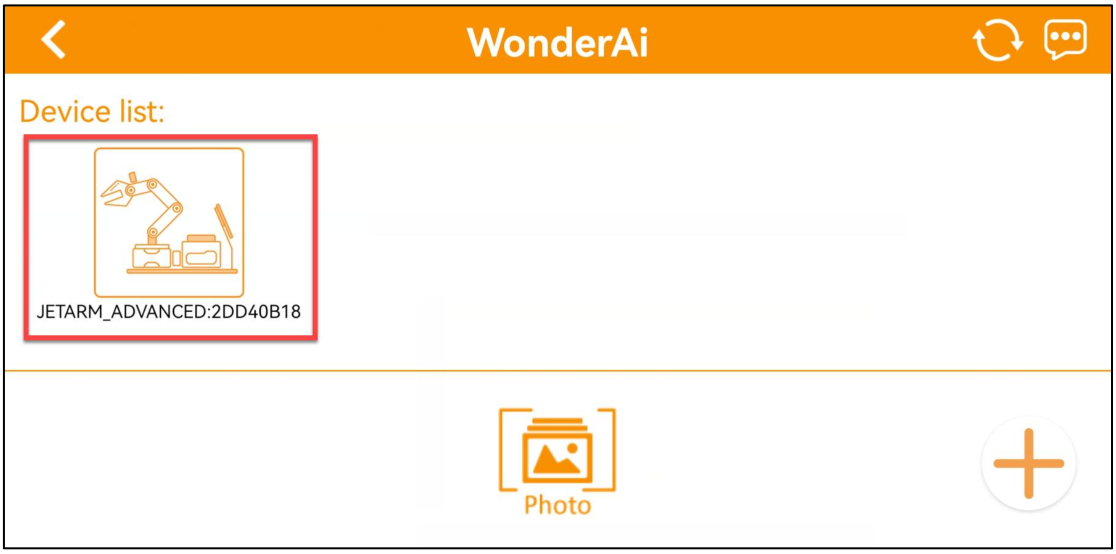


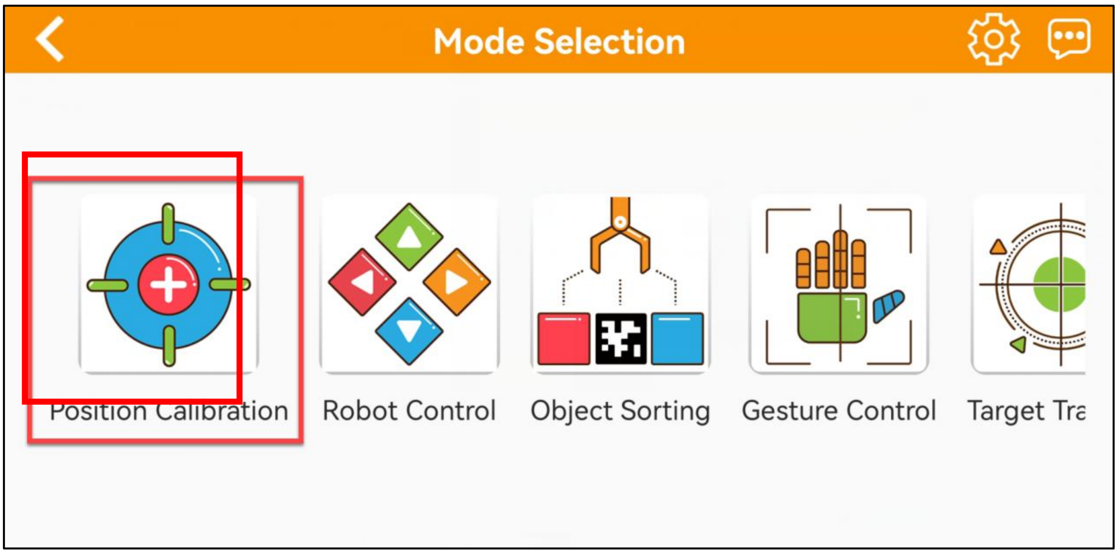
7. Position Calibration

Position calibration realizes the transformation of the coordinates on your phone screen to the actual physical locations of robotic arm, allowing for better execution of the sorting.

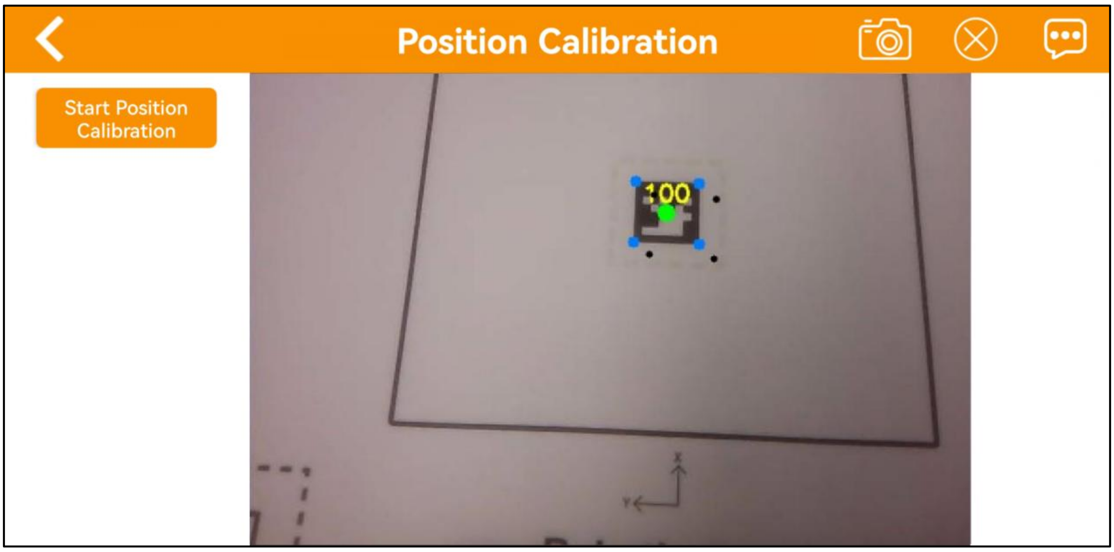
Note: please strictly follow the operation below to calibrate robotic arm's position, do not jump this step.

1) Connect the robotic arm according to "**5.2 Connection Mode Introduction**", then click the robotics icon to enter the mode selection interface, choose "Position Calibration".

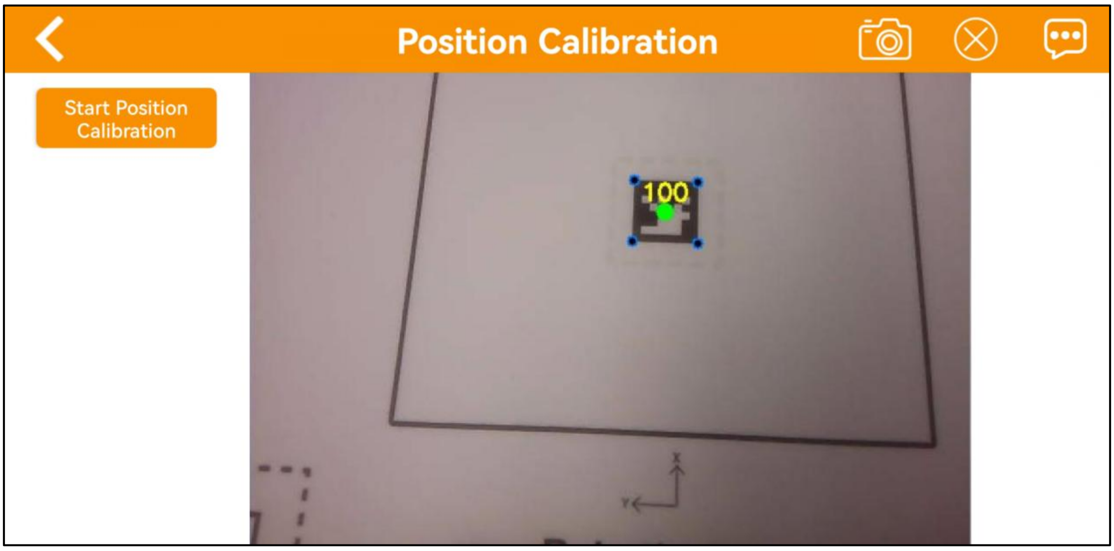




2) When using the app for the first time, the screen on the right may take a few seconds to appear. If there is no live camera feed for a long time, return to the mode selection page and re-enter. Once the live camera appears, tap "**position calibration**".



3) Please be patient. Calibration needs to take a while to complete. Upon observing the alignment of 5 block dots overlap with the 5 colored dots on AprilTag, the calibration is finished.



8. APP Control

Following will specify each function of app. Here takes iOS system as example for demonstration. The operations are application to Android system.

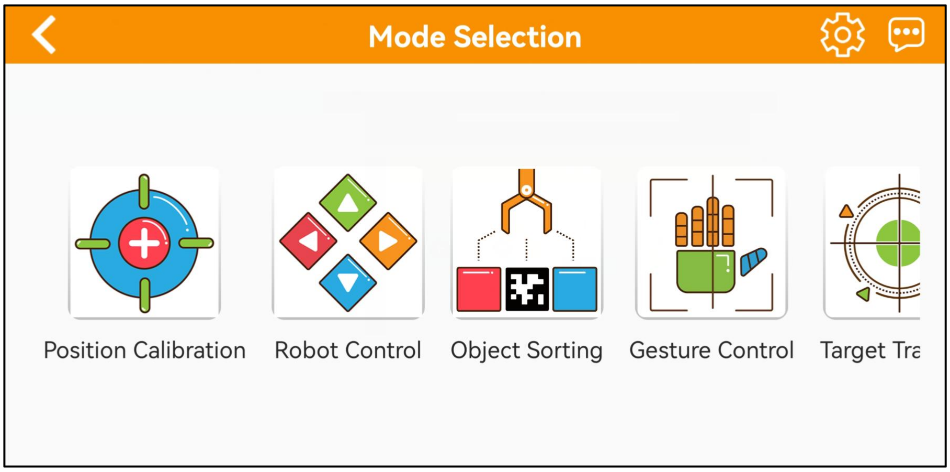
8.1 Preparation

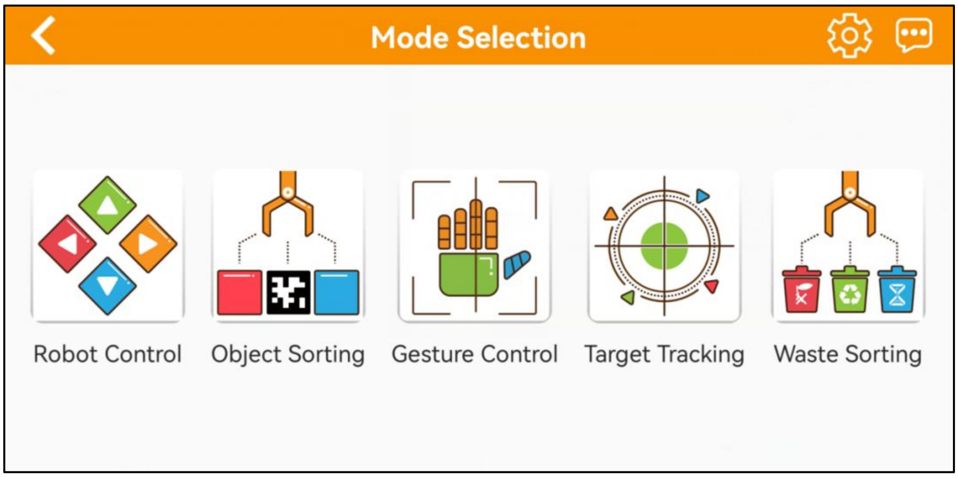
1) Start JetArm. For the details on robot's initial startup, please refer to the content of "**4. Initial Startup and Test**".

2) Connect it to "**WonderAi**" app according to "**5. App Installation and Connection**".

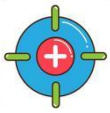
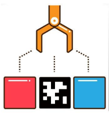
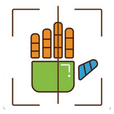
8.2 APP Introduction

Take JetArm ultimate kit as example. There are 6 game modes in app, including position calibration, robot control, object sorting, gesture control, target tracking, waste sorting.





The instructions of each games are as follow:

Icon	Mode	Performance
	Position calibration	Obtain the actual distance between the block and the end-effector of robotic arm.
	Robot Control	Control the motion of robotic arm in real time.
	Object Sorting	After selecting the block or card to track in the app, robotic arm will transport the corresponding block from the recognition area to the sorting area.
	Gesture Control	After starting this function, the camera will look upward. When recognizing the gesture, robotic arm will perform the corresponding action.
	Target Tracking	Select the color or face to track in the app, and the robotic arm will move following the movement of the target.
	Waste Sorting	The robotic arm will grab the waste card from the recognition area, and place it in the corresponding waste classification area.

8.3 APP Operation

8.3.1Position Calibration

Position calibration realizes the transformation of the coordinates on the live camera feed to the physical locations of robotic arm. Though position calibration, the actual distance between the block and robotic arm can be obtained, facilitating grabbing and placing.

8.3.2 Robot Control

All servos on robotic arm can be controlled through this game.

Note:

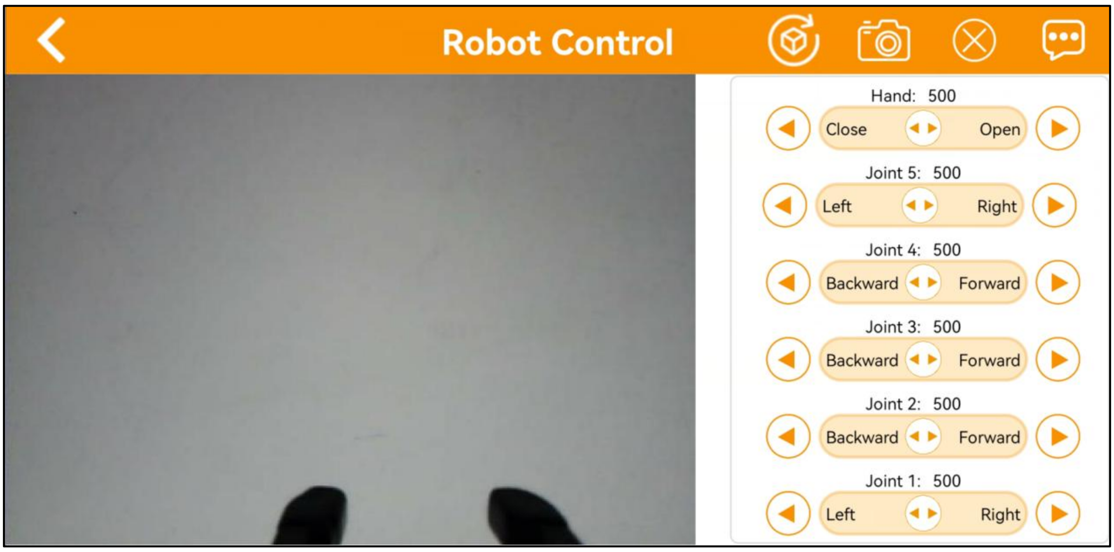
1. Keep your body away from the robotic arm to prevent injury.

2. When robotic arm reach its limit position, please control it in the opposite direction.

- **Interface Layout**

1）Click  to enter the game interface, then robotic arm returns to an straight posture.

2）The interface of "Robot Control" consists of two parts. The left side of the interface is the live camera feed. The right side is the robotic arm control area.



- **Function Instruction**

Click the control buttons at right to control the corresponding servos.

Icon	Instruction
	Click the corresponding buttons to respectively control the motion of joints 1-5 as, well as the hand joint.
	Click it to get all servo back to their central posture (The robotic arm presents straight posture).
	Capture the current camera view
	Close the navigation bar. Tap the black to show it again.
	Display the related information of Hiwonder

8.3.3 Object Sorting

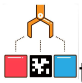
Note:

1. Please start this game in a well-lit environment.

2. Keep a certain distance between the blocks. Do not make them too close to each other.

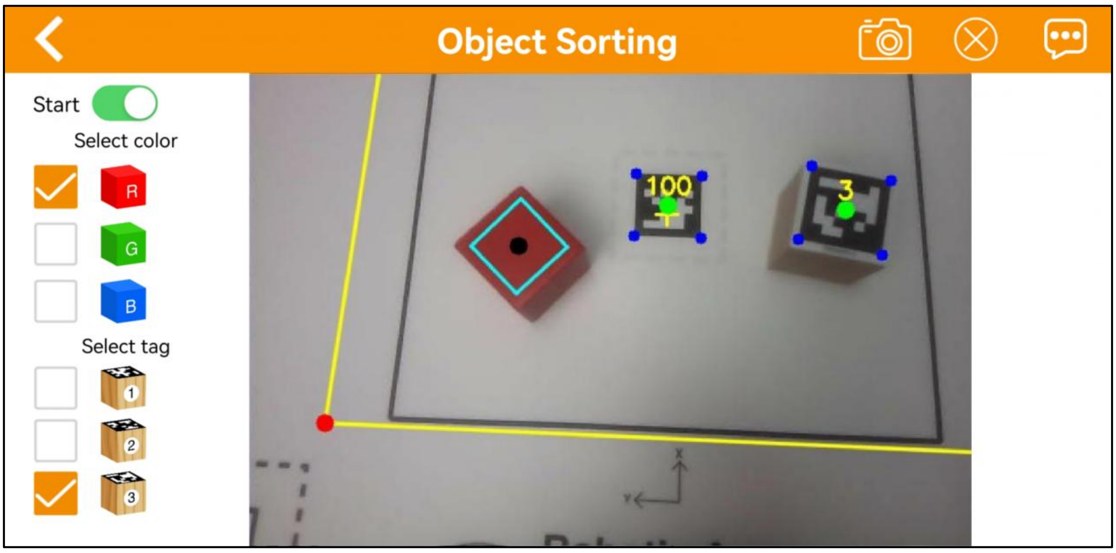
3. Maintain the integrity of the tag as the missing corners or stains may effect the recognition result.

- **Interface Layout**

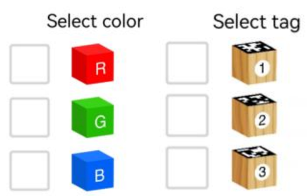
Click  to enter the game interface. The interface consists of two parts.

① On the left side, you can start the game and select the target color and tag.

② The right side is the live camera feed.



- **Function Instruction**

Icon	Instruction
	Activate/Deactivate game
	Select the color and tag for sorting
	Capture the current camera view

	Help information

- **Operations and Outcome**

After selecting the color and tag for sorting on the left, activate the game.

When the camera recognizes the target block within the sorting area, it will pick it up and move it to the designated corresponding position.

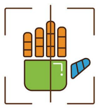
8.3.4 Gesture Control

Note:

1. The robot can only recognize one hand at a time.

2. The time interval for each gesture recognition is 5 seconds.

- **Interface Layout**

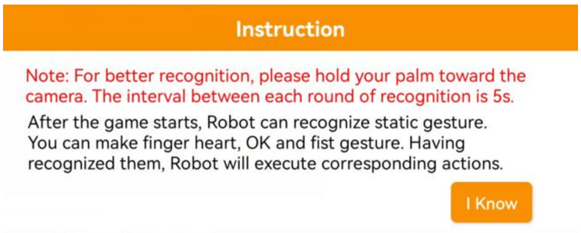
Click  to enter the game interface. The interface is consists of two parts.

1) On the left side, you can start the game.

2) The right side is the live camera feed.

- **Function Instructions**

After the game starts, the robotic arm rotates horizontally. When the camera detects a human face, the robotic claw will perform a rotating and gripping motion.

Icon	Instruction
Start	Activate/Deactivate game
Instruction	Click it to obtain operation instruction 
	Capture the current camera view
	● Close the navigation bar
	Help information

- **Operation and Outcome**

Gesture control refers to the static gesture recognition. Once the robotic arm recognizes the gesture within the camera's field of view, it responds with corresponding actions.

- **Static Gesture Recognition**

Recognizable static gestures and its corresponding actions are as follow.

Static Gesture	Example	Action
Hand heart gesture		The pan-tilt turns right, and the robotic claw rotates to the right. The pan-tilt turns left, the robotic claw rotates to the right.
OK		Wave its arm
Clenching fist		Robotic arm moves left and right after gripping
"Gun" gesture		Robotic arm patrols toward the right
1		Gripping and placing

2		Grip and place it in the upper right corner
3		Grip and place it in the right front.
4		Grip and place in front
5		The robotic claw opens, and the pan-tilt turns left and right.
6		The robotic claw closes, and the pan-tilt turns left and right.

8.3.5 Target Tracking

Note:

1. Please start this game in a well-lit environment. After the game starts, please remove other objects in target color from the field of view of the camera, otherwise the game will be influenced.

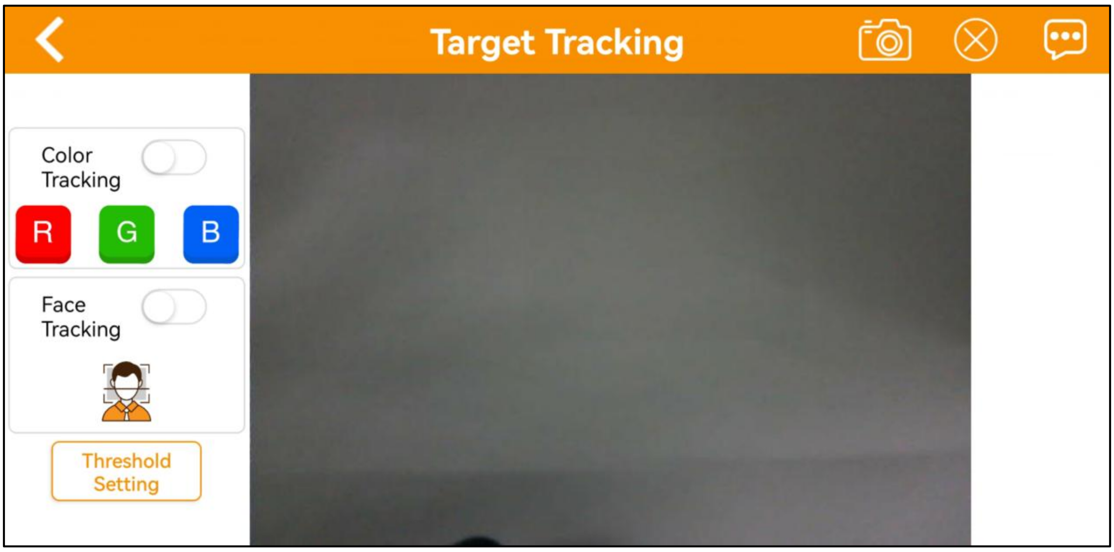
2. The target must always locate within camera's view, and move it in an appropriate speed.

- **Interface Layout**

Click  to enter the game interface. The interface is consists of two parts.

1) On the left side, you can start the game and select the target color.

2) The right side is the live camera feed.



- **Function Instruction**

Icon	Instruction
	Activate/deactivate game

R G B	Select the target color
Face Tracking 	Activate/deactivate face tracking
Threshold Setting	Tap it to enter the color threshold adjustment interface (For the adjustment operations, please refer to “9. Color Threshold Adjustment”).
Instruction	Game operation instructions
	Capture the current camera view
	Help information

- **Operations and Outcome**

After the color tracking starts, it is necessary to select a target color. You can slowly move the object in front of camera. When the camera detects the object, the robotic claw will move with it.

8.3.6 Waste Sorting

Note:

1. Please start this game in a well-lit environment.

2. For accurate gripping, keeping the block with waste card parallel to the lines of the gripping area.

---

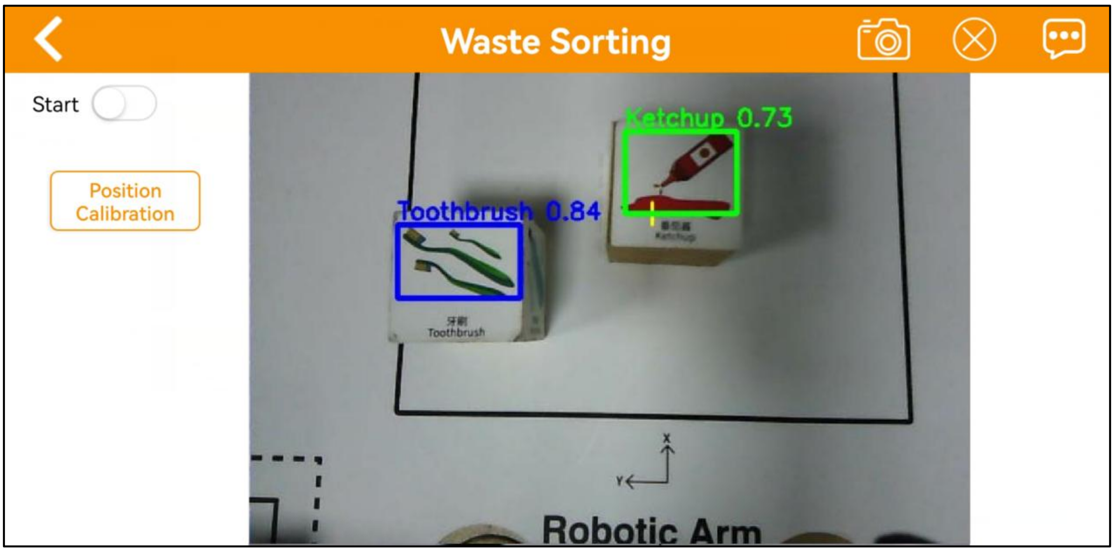
3. If the gripping is inaccurate, you can tap the position calibration button to perform calibration.

Interface Layout

Click  to enter the game interface. The interface is consists of two parts.

3) On the left side, you can start the game.

4) The right side is the live camera feed.



- **Function Instructions**

Icon	Instruction
	Activate/deactivate game
	Tap this button to enter the position calibration interface (For the detailed operations, please refer to “7. Position calibration”.
	Capture the current screen view

	Help information

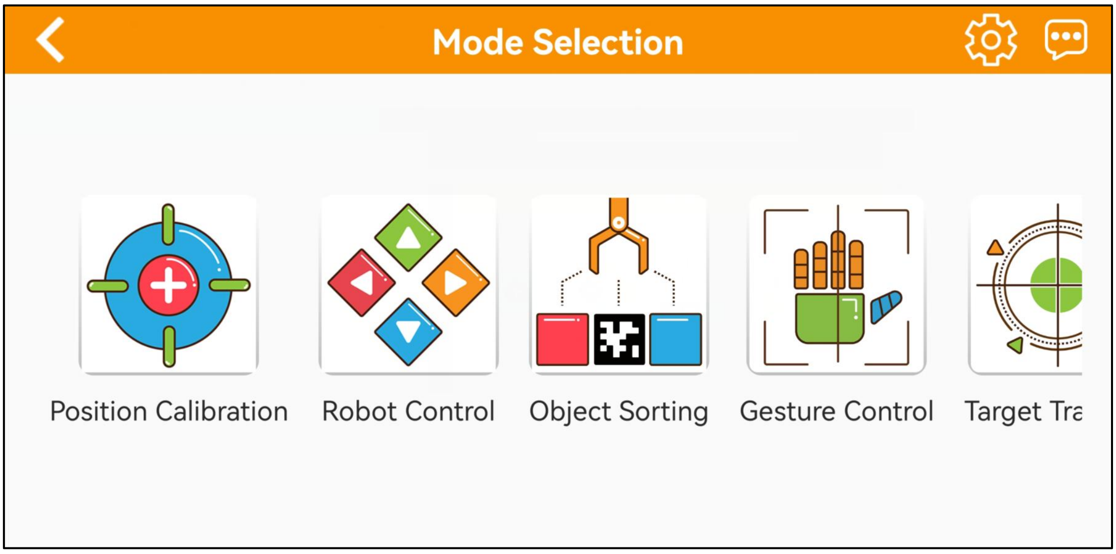
- **Operations and Outcome**

Place the bock with waste card within the recognition area. When the camera recognizes a block, the robotic arm will sequentially stack the blocks by category into their respective areas

9. Color Threshold Setting

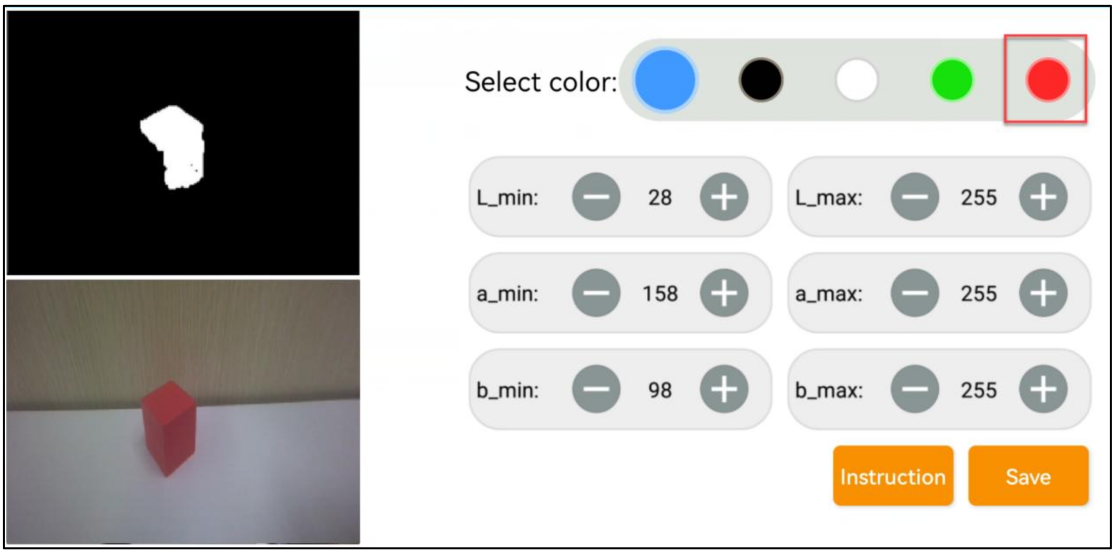
When we playing the AI vision games, the environment light will affect the JetArm’s performance. And we can adjust the color threshold to cope with this. Take Android system as example for demonstration. It is also applicable to iOS system.

1) Start JetArm and connect it to “**WonderAi**”. (For the operation instruction, please refer to “4. Initial Startup” and “5. App Installation and Connection”.)

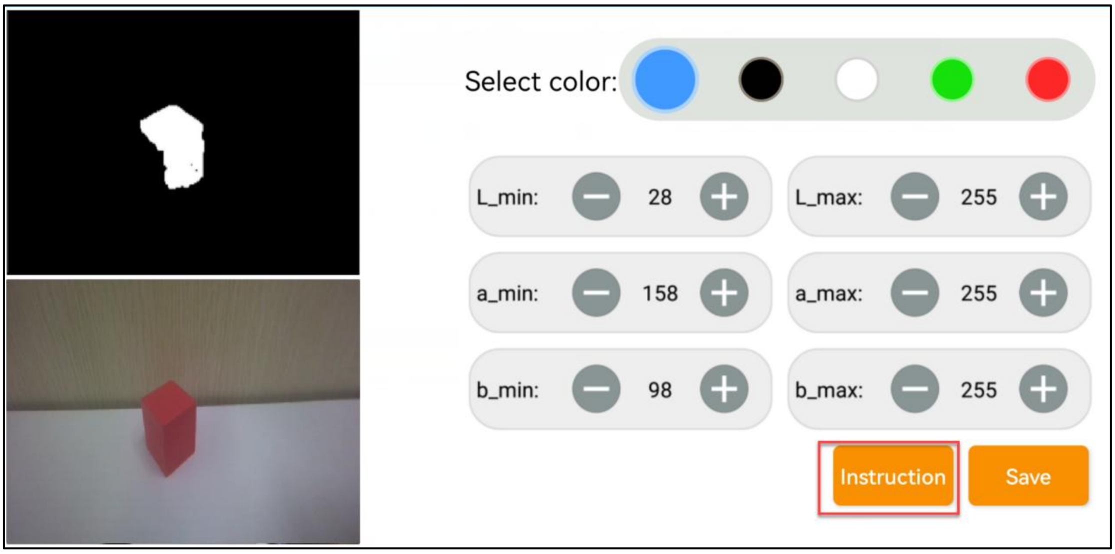


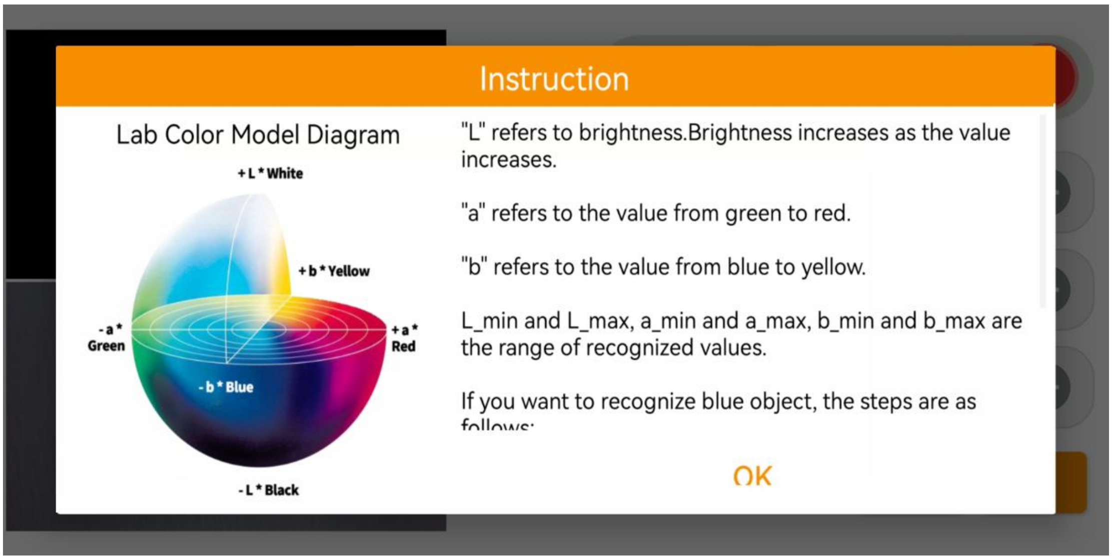
2) Enter the main interface, then click “” at the upper right corner to enter the interface of color threshold setting.

3) Take adjusting red color for example. Place the red block within the recognition area on the map, then select the red icon.



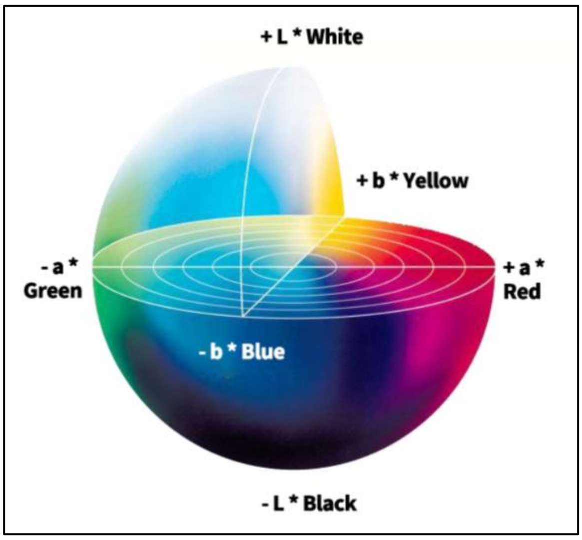
4) Click “**Instruction**” button to check the color adjustment instruction.



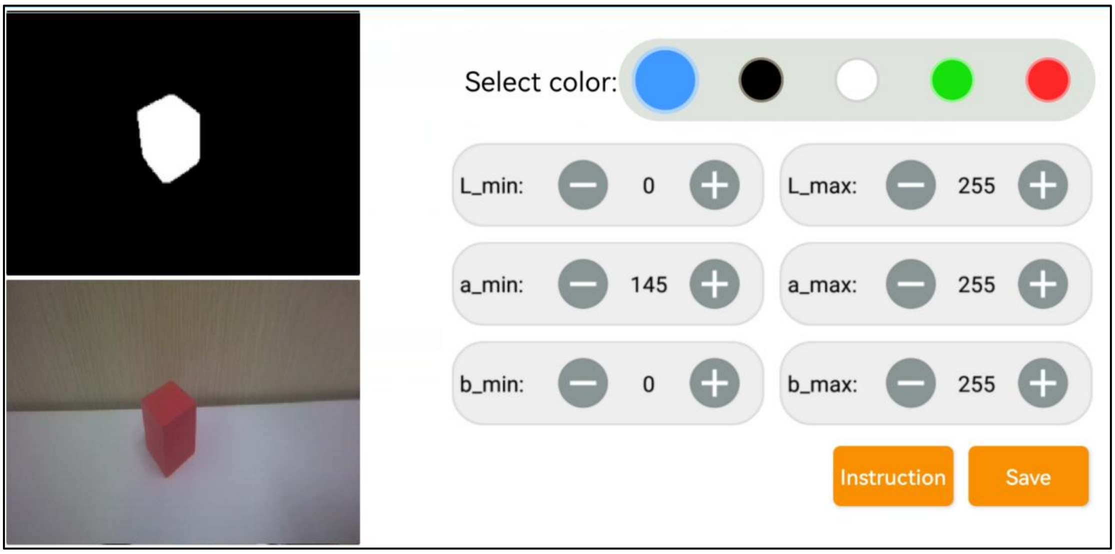


5) Set the parameters of **L_min, a_min and b_min** as 0, and the parameters of **L_max, a_max and b_max** as 255.

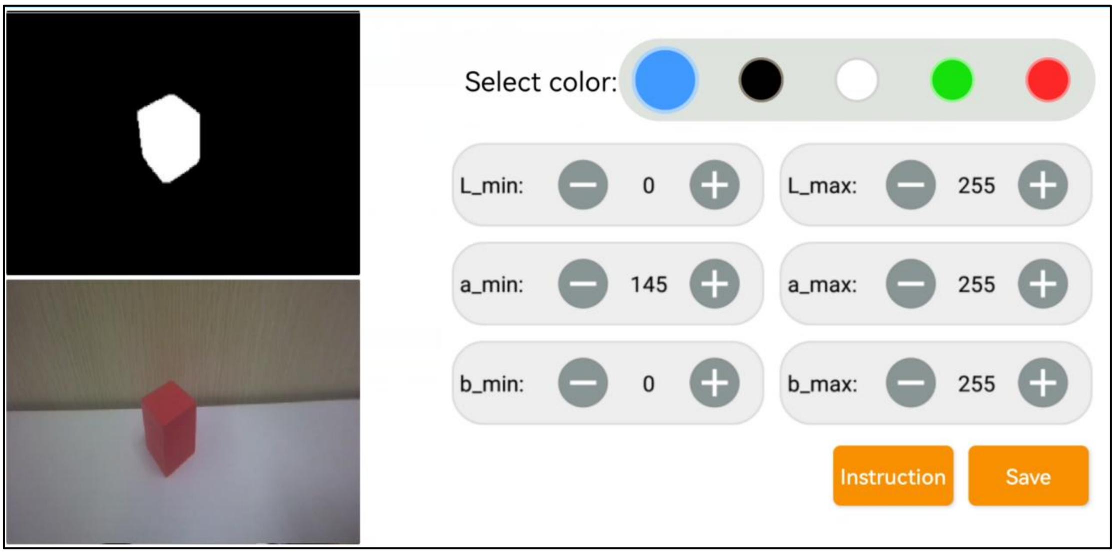
6) Refer to LAB color space distribution to adjust the L, A and B components.



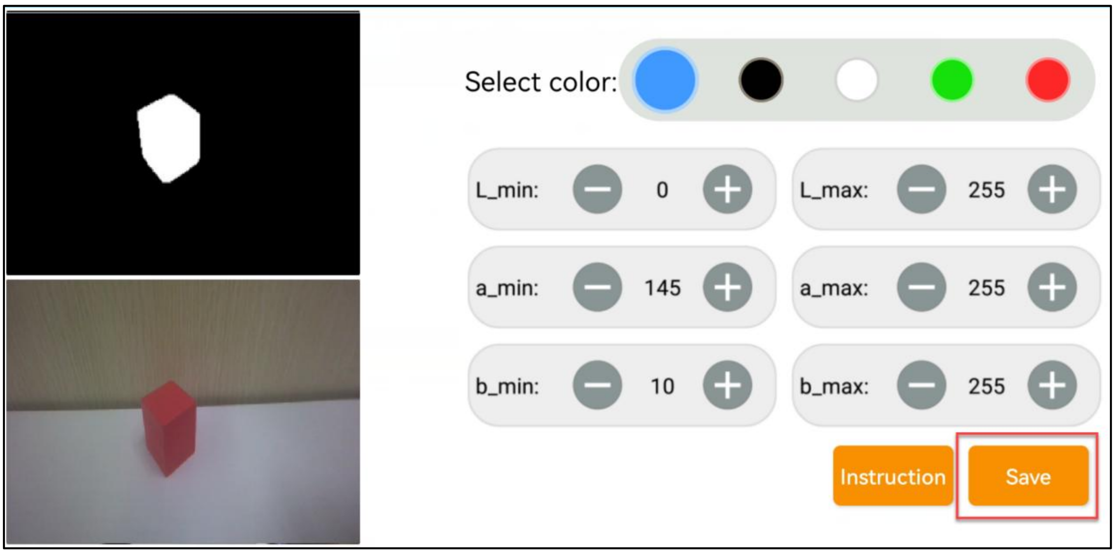
7) According to the Lab color model diagram, red color is around “**+a**” zone, so we need to adjust the color threshold to “**+a**” zone. Keep the values of “L_**max**” unchanged, while increase the values of “L_**min**” until the block image in the screen turns white and other area turns black.



8) Then based on the environment, adjust the value of **L_min, L_max, b_min and b_max.** If the color belongs to light red, you need to increase the value of “**L_min**”. If it belongs to dark red, decrease the value of “**L_max**”. If it belongs to dark red, decrease the value of “**L_max**”. If it belongs to warm tone, increase the value of “**b_min**”. If it belongs to cool tone, decrease the value of “**b_max**”



9) After the adjustment is completed, tap the “Save” button to save the color threshold.



10. Wireless Handle Control

10.1 Preparation

1) Insert the handle receiver into any USB interface on Jetson Nano.

Note: please insert the handle receiver before starting the robot.

---

2) Please bring your own two AAA dry batteries. And insert them into the battery slot.And do not invert the positive and negative poles.

10.2 Device Connection

1) Start JetArm.

2) Switch on handle. At this point, two LEDs (red and green) on the handles will flash simultaneously.

3) Please wait for a while. Then the robot will pair with the handle automatically. After successful pairing, the green light will keep lighting up and red LED light goes out.

Sleeping mode: If the handle doesn't connect to the robot within 30s or there is no operation on the handle within 5 minutes after turning on, it will enter sleep mode. And you can press "START" to activate the handle.

10.3 Control Mode Introduction

The handle control for the robotic arm are divided into two modes: coordinate mode and single servo mode. When the handle is successfully connected, it defaults to the coordinate mode.

Single servo mode: by pressing the buttons on handle, you can control individual servos on the robotic arm for forward and reverse movements.
Coordinate mode: by pressing the buttons on handle, you can control the robotic arm as a whole along the trajectory of three axes (X, Y, Z) and combined with angular deviation for motion.

The way to switch between the two modes: press the "SELECT" and "START" buttons. Once you hear a beep sound, the mode has been successfully switched. Switching from single servo mode to coordinate mode, you will be prompted through two beeps sound. One beep sound represents the switch from coordinate mode to single servo mode.

10.4 Button Function

10.4.1 Single Servo Mode

Following is the button function instruction for single servo mode:

Button	Function (from the first-person perspective of robotic arm)
START	Return robotic arm to its initial posture
SELECT+START	Switch the servo control modes (singe servo mode/coordinate mode)
UP / ↑	Raise servo No.2
DOWN / ↓	Lower servo No.2
LEFT / ←	Rotate Servo No.1 to the left
RIGHT / →	Rotate Servo No.1 to the right
TRIANGLE / Δ	Close robotic claw (servo No.10)
CROSS / X	Open robotic claw (servo No.10)
CIRCLE / ○	Rotate No.5 servo to the right (the robotic claw turns right)
SQUARE / □	Rotate No.5 servo to the left (the robotic claw turns left)
L1	Raise servo No.3

L2	Lower servo No.3
R1	Raise servo No.4
R2	Lower servo No.4

10.4.2 Coordinate Mode

Following is the button function instruction for coordinate mode:

Button	Function (from the first-person perspective of robotic arm)
START	Return robotic arm to its initial posture
SELECT+START	Switch the servo control modes (singe servo mode/coordinate mode)
UP / ↑	Robotic arm moves along the positive direction of X-axis (the front of robotic arm)
DOWN / ↓	Robotic arm moves backward along the negative direction of X-axis
LEFT / ←	Robotic arm moves along the negative direction of X-axis (robotic arm's left side)
RIGHT / →	Robotic arm moves along the negative direction of Y-axis (robotic arm's right side)
TRIANGLE / Δ	The robotic gripper closes (ID10 servo)
CROSS / X	The robotic gripper opens(ID10 servo)

CIRCLE / ○	The ID5 servo turns right (Robotic gripper turns right)
SQUARE / □	The ID5 servo turns left (Robotic gripper turns left)
L1	Robotic arm moves along the positive direction of Z-axis (above the robotic arm)
L2	Robotic arm moves along the negative direction of Z-axis (under the robotic arm)
R1	Increase the reflection angle (based on the horizontal line)
R2	decrease the reflection angle (based on the horizontal line)

11. Remote Desktop Installation and Connection

Following is a general tutorial. The device name and image appeared in this section is only for you reference, and please subject to the actual device that you're using.

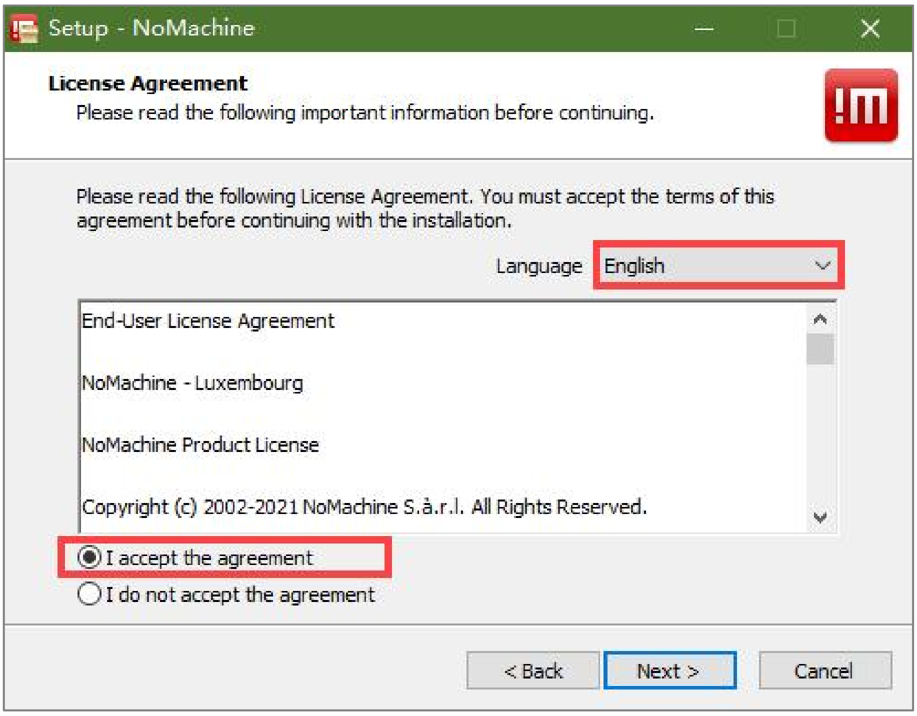
11.1 NoMachine Installation

The installation operations are as follow:

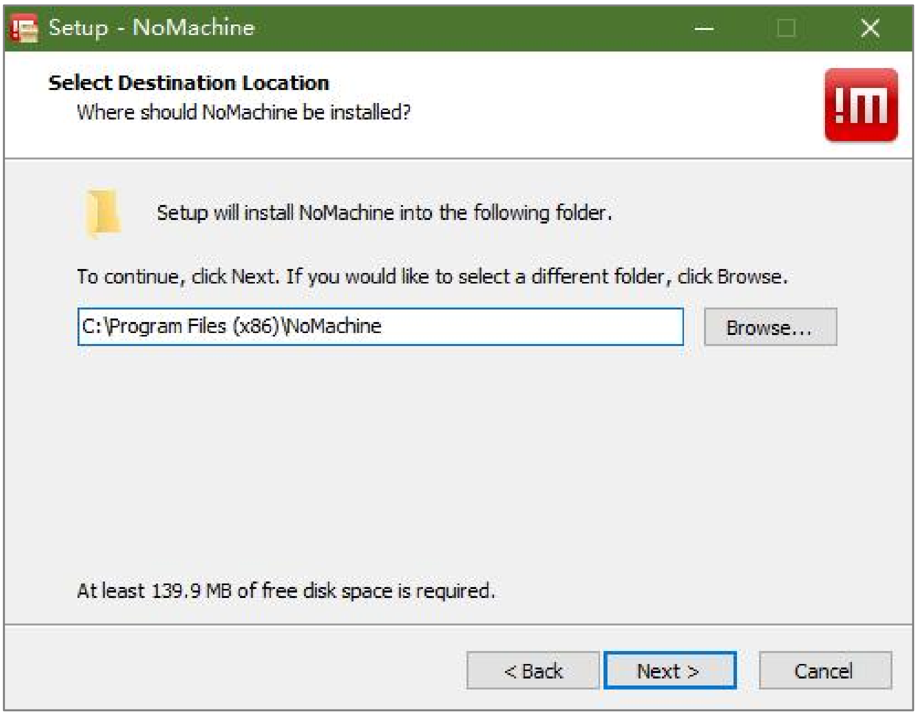
1） Open the software pack "**nomachine_8.4.2_10_x64**" in the "software" folder.

2） Click “**Next**” button.

3） Set the language as “**English**” and tick “**I accept the agreement**”. Then move to “**Next**” step.

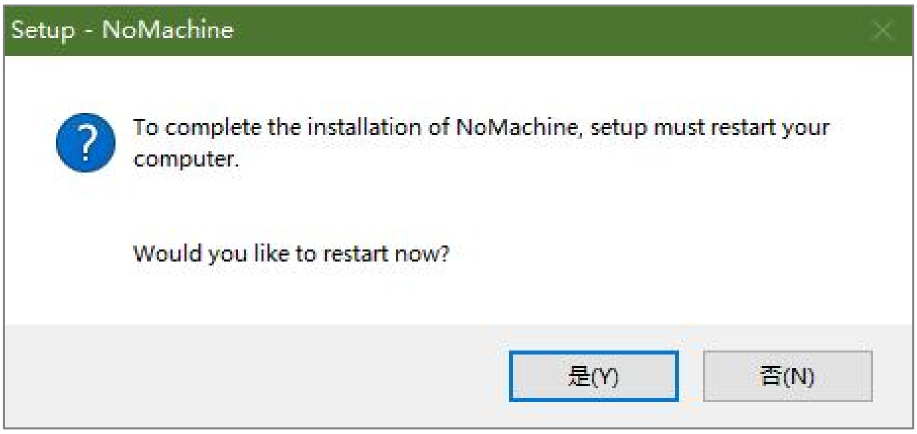


4） Remain the default storage path for NoMachine, then click “**Next**”.



5） Please wait until the installation completes. Click "**Finish**" button.

6） Click "**Yes**" to restart the computer. Please don't skip this step!



11.2 Device Connection

Note：When establishing a remote desktop connection, JetArm kit with a screen must disconnect the screen power cable before booting up. Otherwise, it may cause resolution issues.

You can remotely connect to the device in three ways as follows:

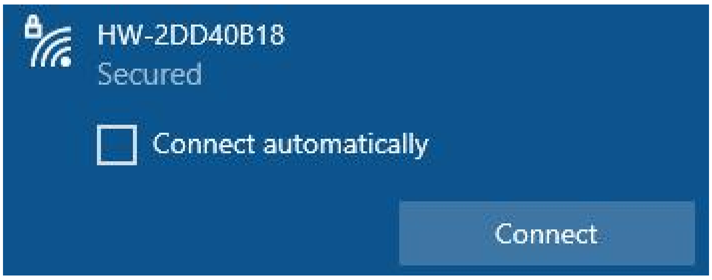
① AP direct connection mode: the demo board can generate a WiFi network that can be accessed by a phone. However, in this mode, the user cannot connect to the Internet.

② STA LAN mode: the demo board can connect to the specific hotspot/ Wi-Fi. Different from AP direct connection mode, you can access the external Internet in STA LAN mode.

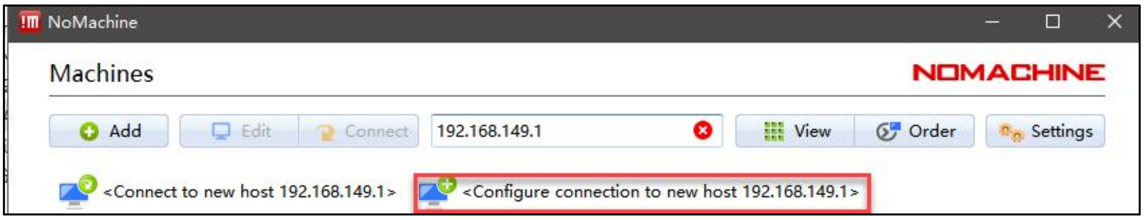
③ USB connection: connect the demo board and computer through USB cable, then remotely access the robot’s system desktop using the fixed IP address. However, in this mode, the user cannot connect to the Internet.

11.2.1 Access under AP Direct Connection/ STA LAN Mode

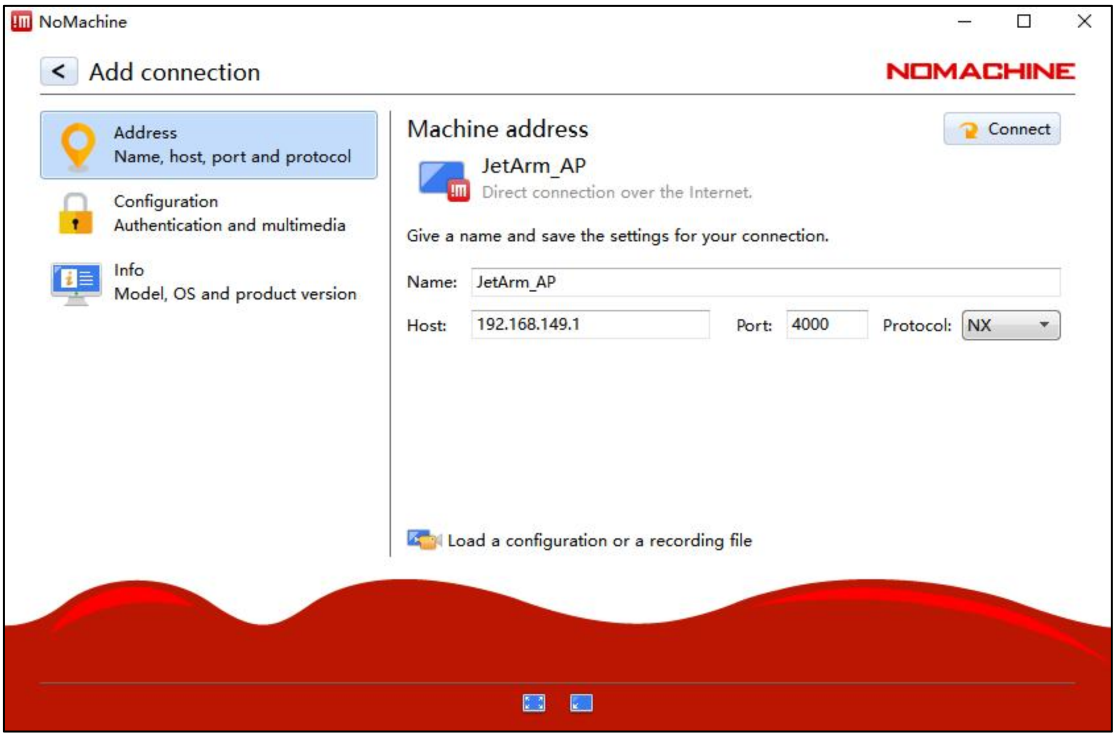
JetArm is default to AP direct connection mode. After it boots up successfully, it will create a WiFi starting with ‘**HW**’. Connect your computer to this WiFi.



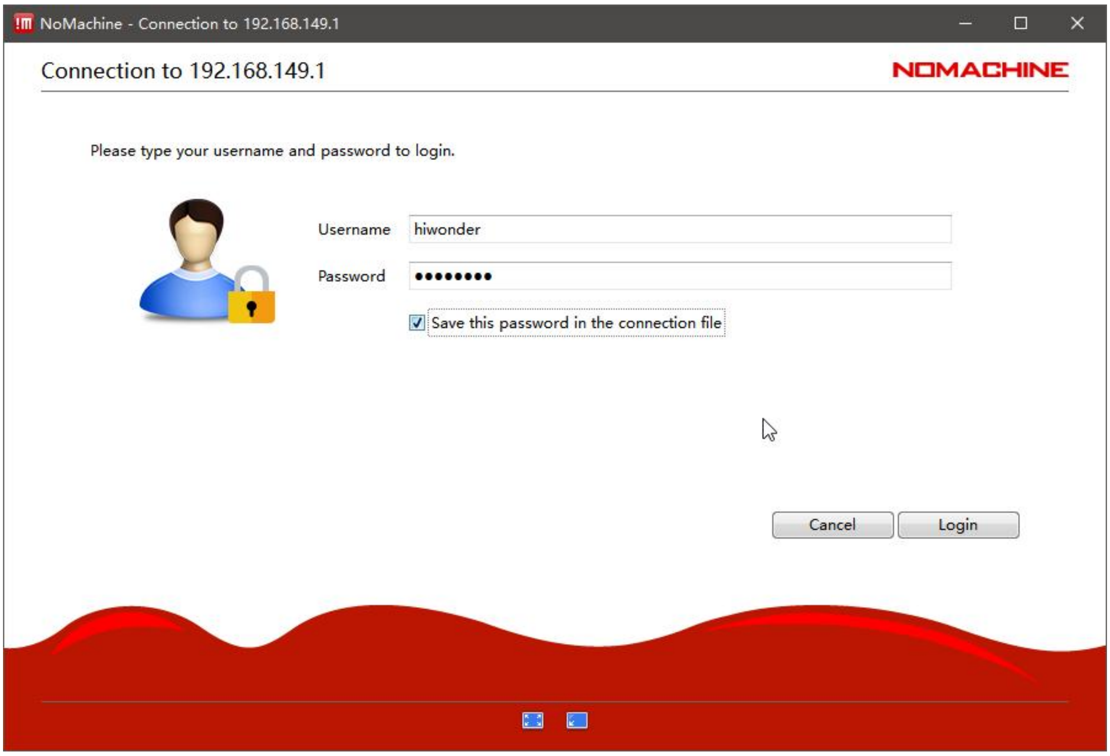
2) Open NoMachine. Input “**192.168.149.1**” in the search bar, then click-on “**Configure connection to new host 192.168.149.1**”.



3) Change the name to “**JetAuto**”, then click-on “**Connect**” button.

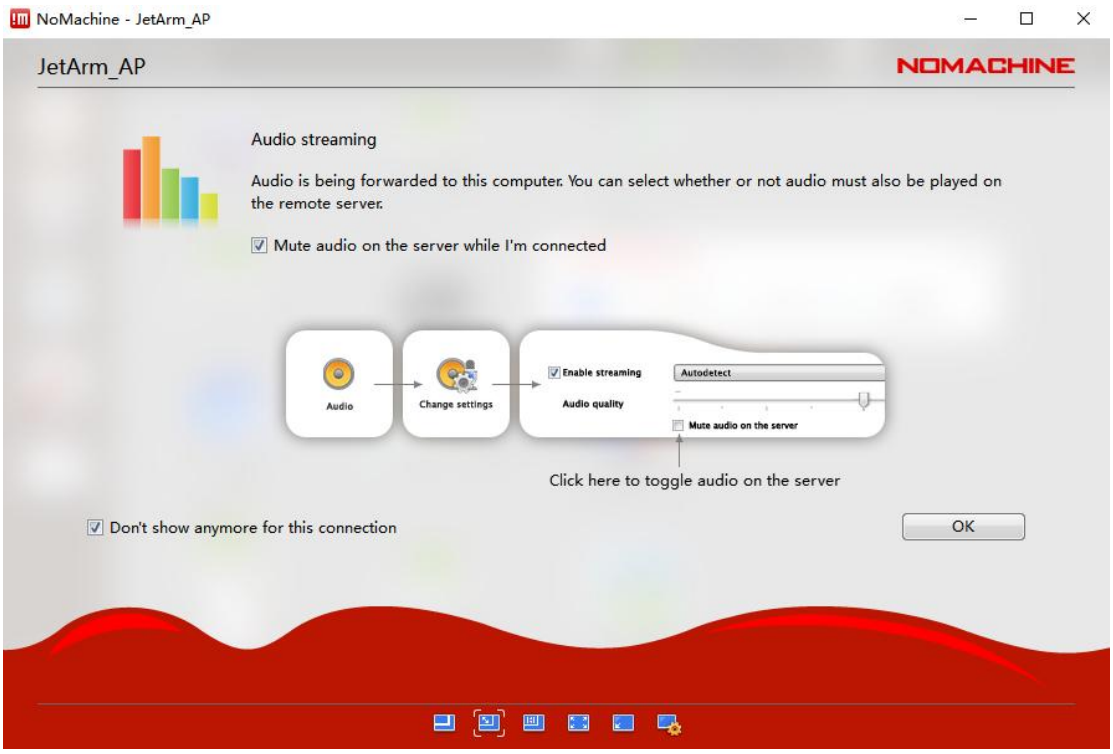


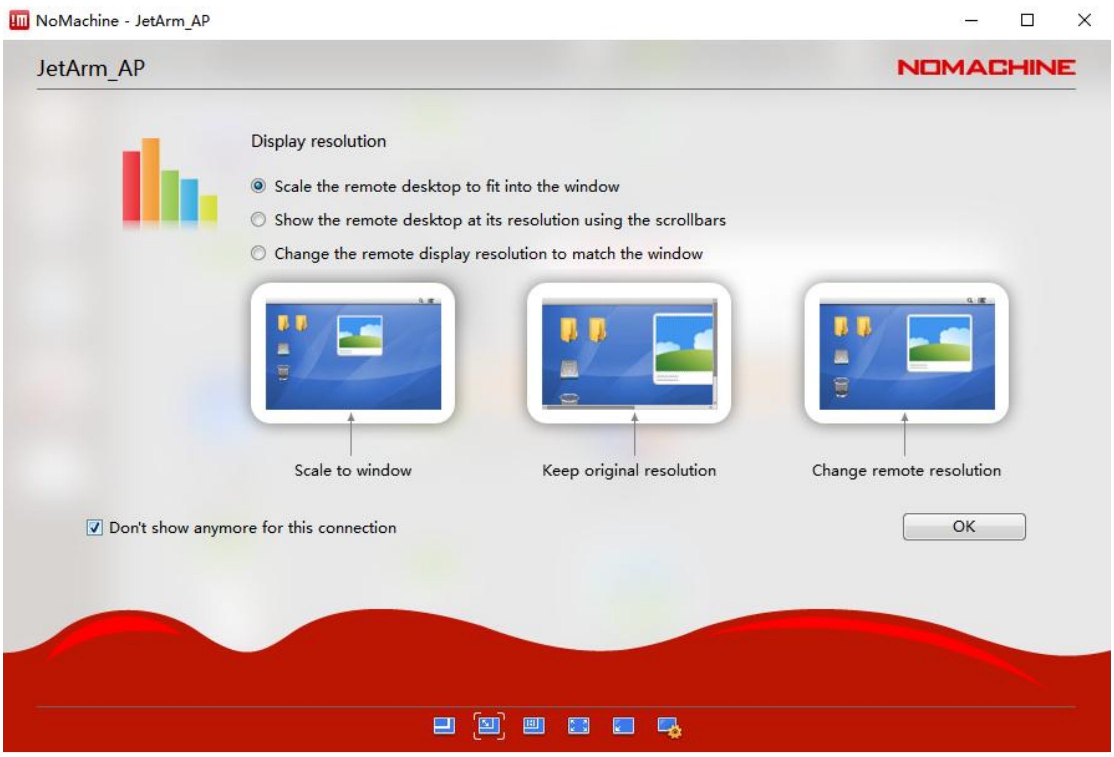
4) Use the username **"hiwonder"** and password **"hiwonder**" to log in. Remember to select "**Save this password in the connection file**" before clicking the "**Login**" button to access the Jetson Nano desktop.

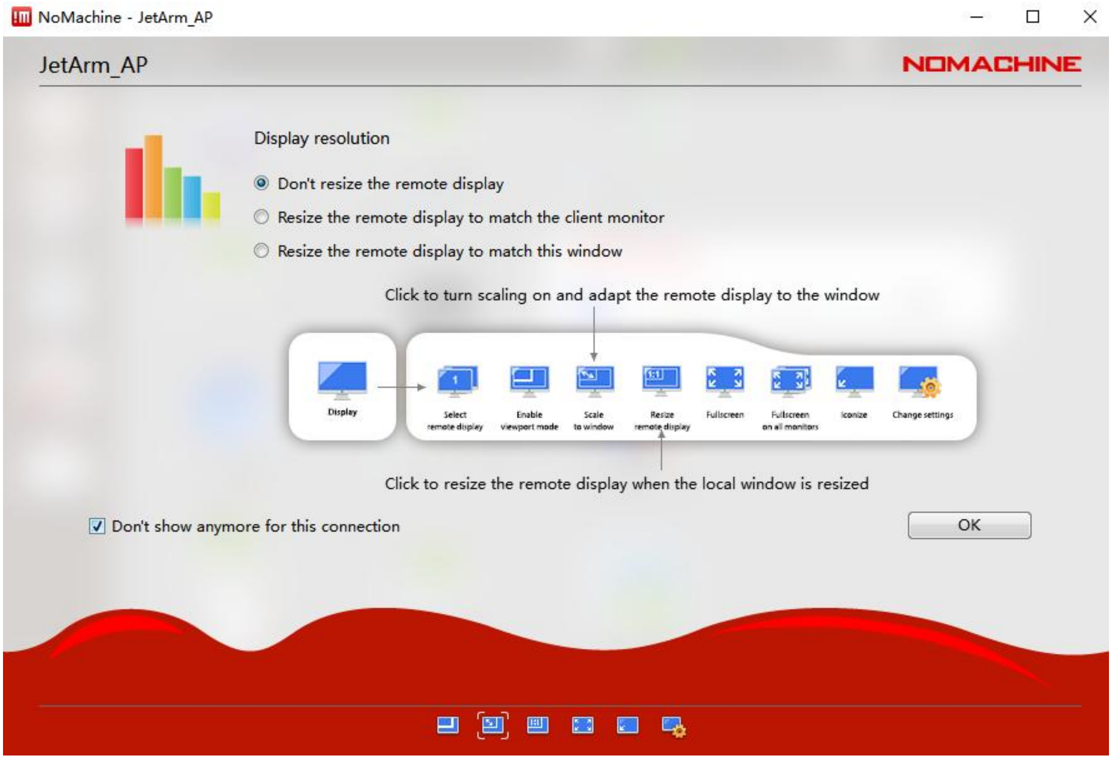


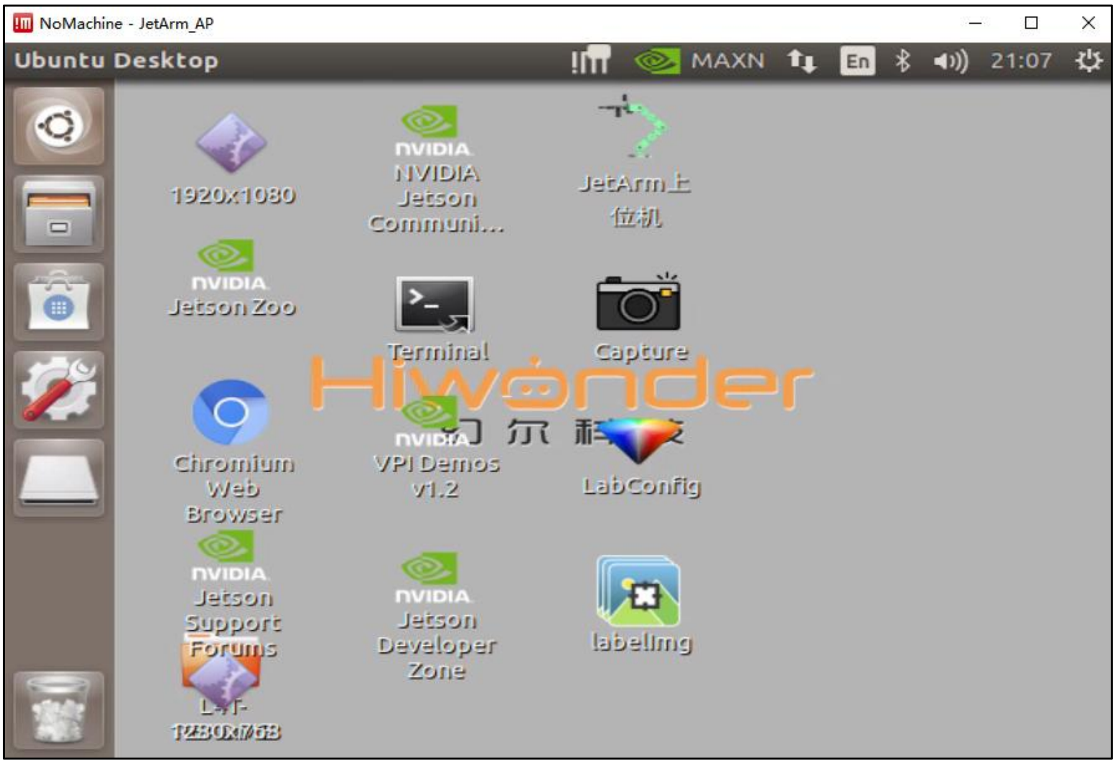
Note: if the robot is configured as STA LAN mode, above operations are also applicable. You need to replace with the corresponding IP, username (it can be changed as requirement) and password (hiwonder). (The IP in LAN mode can be viewed in “5.2.2 LAN Mode Connection (Optional)”)

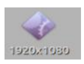
5) Tick the below box and click “Ok”. (If you do not tick this option, the pop-up window will occur next time.)


NoMachine - JetArm_AP
JetArm_AP
NOMACHINE
Audio streaming
Audio is being forwarded to this computer. You can select whether or not audio must also be played on the remote server.
Mute audio on the server while I'm connected
Audio
Change settings
Enable streaming
Autodetect
Audio quality
Mute audio on the server
Click here to toggle audio on the server
Don't show anymore for this connection
OK

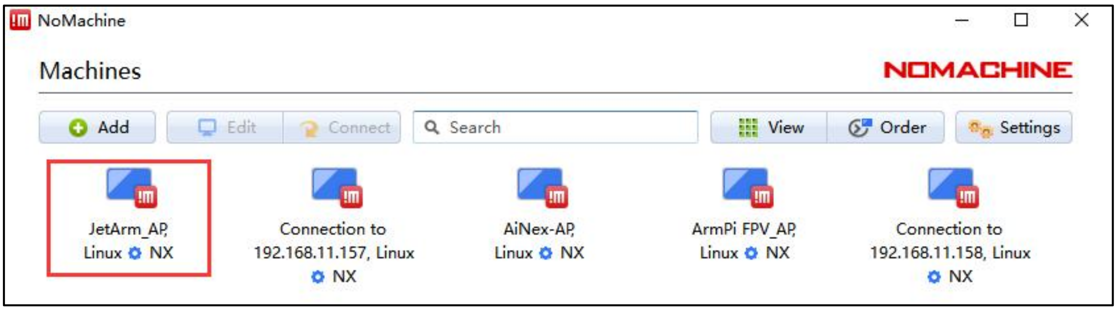

NoMachine - JetArm_AP
JetArm_AP
NOMACHINE
Display resolution
Scale the remote desktop to fit into the window
Show the remote desktop at its resolution using the scrollbars
Change the remote display resolution to match the window
Scale to window
Keep original resolution
Change remote resolution
Don't show anymore for this connection
OK

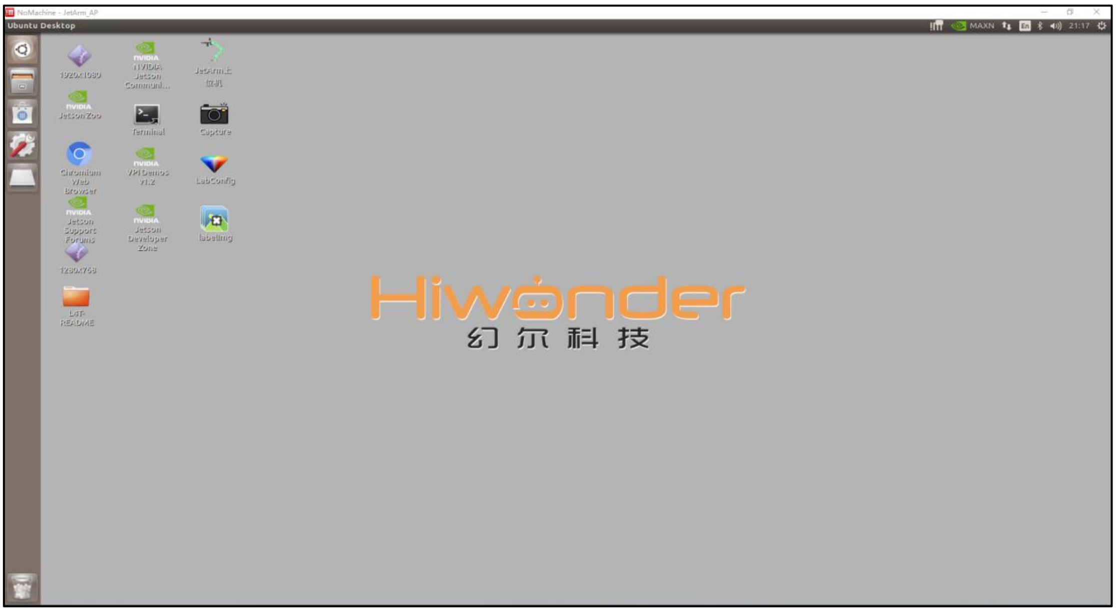




6) After starting up and connecting for the first time, the resolution may be abnormal. At this time, you need to click  on the upper left corner to modify the resolution manually.

7) Then click on the upper right corner to close the current page, then re-connect robot.

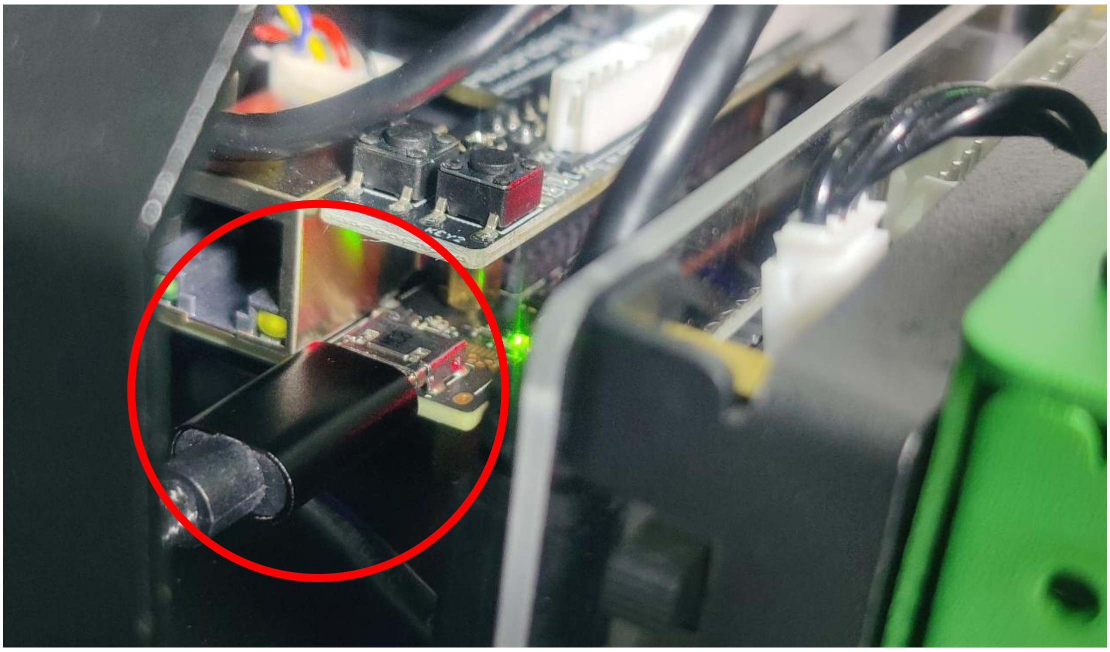




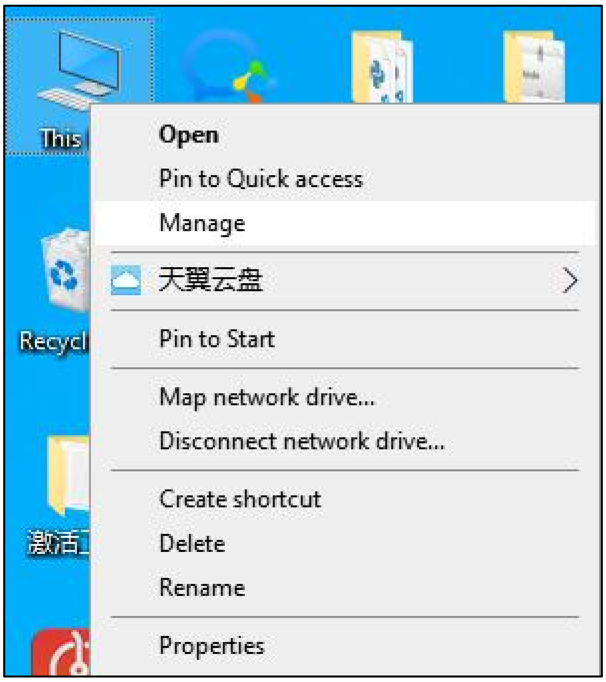
11.2.2 Access via USB cable Using the Fixed IP

If you're using a desktop computer, you can improve the smoothness for remote operations by enabling Remote NIDS compatible device, using the method of the fixed IP "192.168.55.1". The specific operation are as follow:

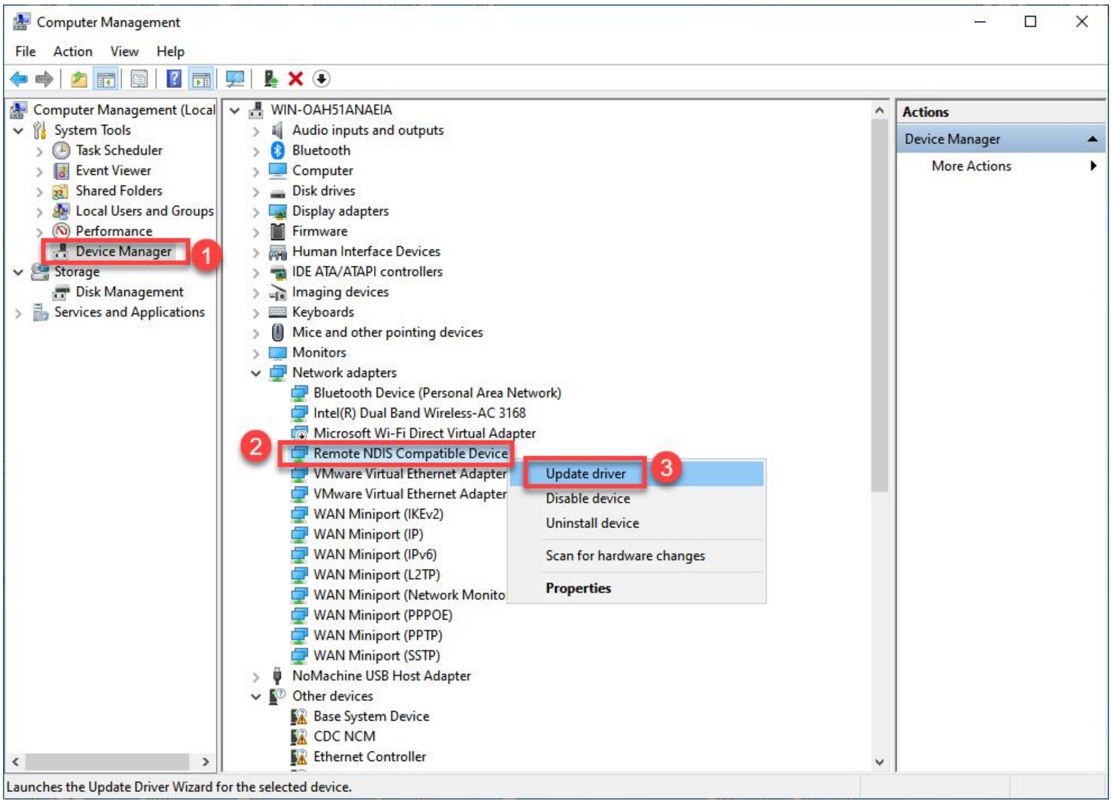
1) Connect your robot to computer with Micro USB cable.



2) Right-click "**This computer**" to choose "**Management**".



3) Click on “**Device Manager”** in the left-hand menu of the interface. Locate the “NDIS driver” under “Network adapters” and right-click on the driver. Choose the “Update driver” option from the menu.

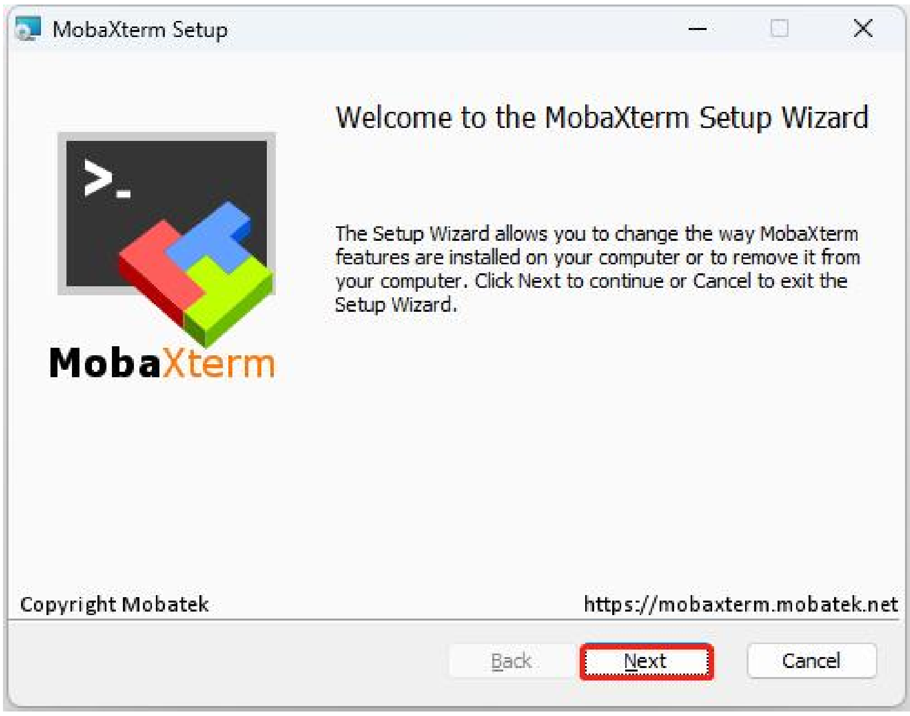


4) After the installation of driver, refer to “11.2.1 AP Direct Connection/ STA LAN Mode Access” to login the system. In addition, you need to replace the IP address with 192.168.55.1, and its password is hiwonder, and username can be modified.

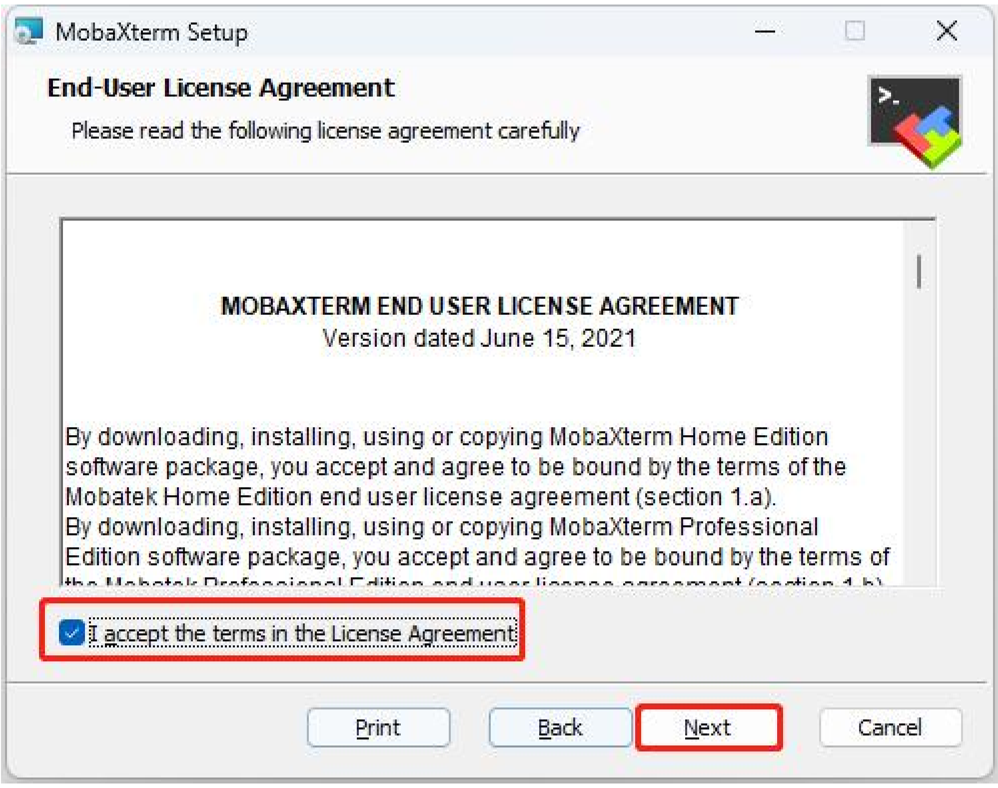
11.3 MobaXterm Installation

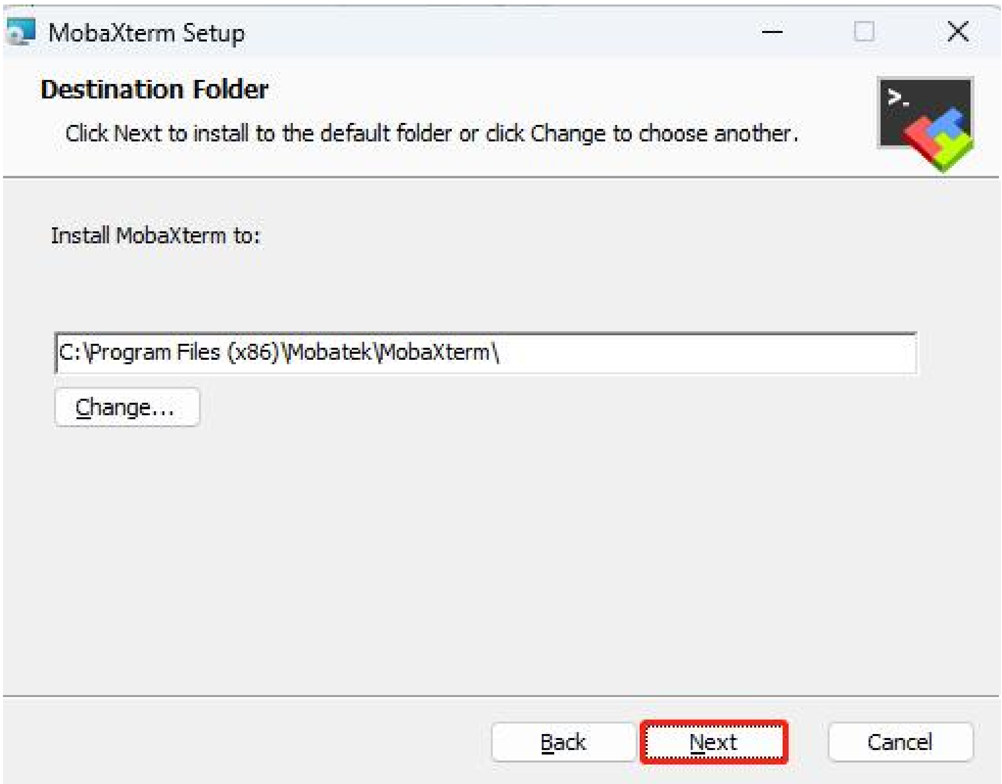
MobaXterm is a software that integrates multiple tools and offers a graphical interface for remote control. By connecting to the Raspberry Pi's WiFi network, you can manage and control it from your computer.

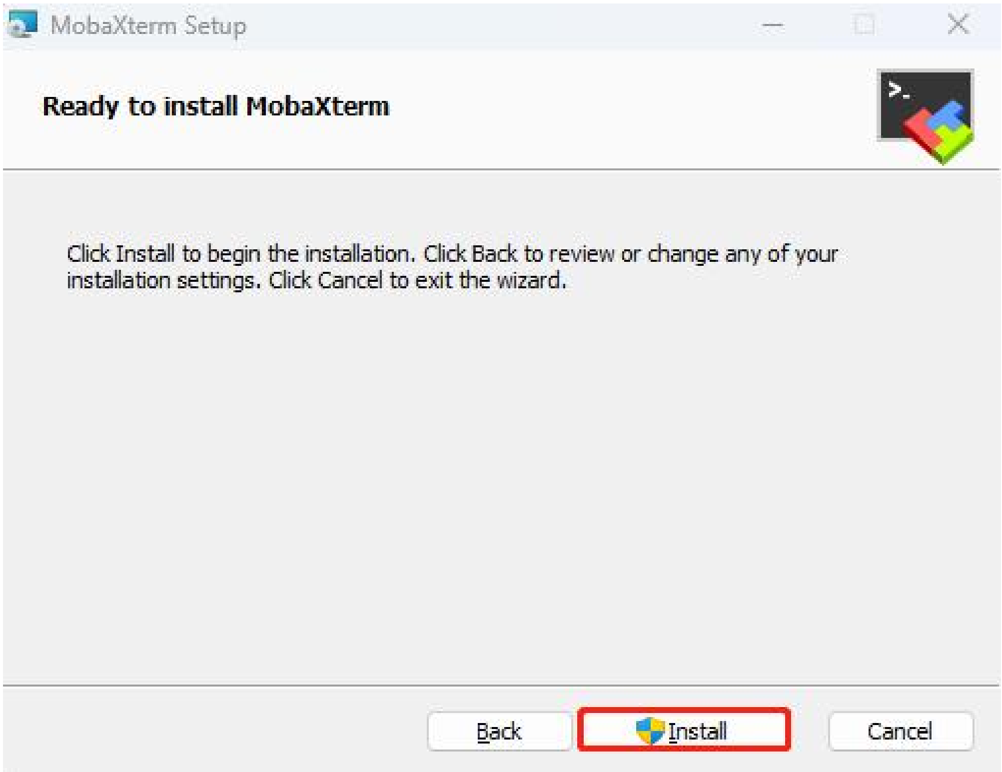
1) Locate the “**MobaXterm_Installer_v22.1**” file in “2. Software/ 2.Remote Desktopn Tool/ MobaXterm”, and double-click the file “**MobaXterm_installer_22.1.msi**” to install it.

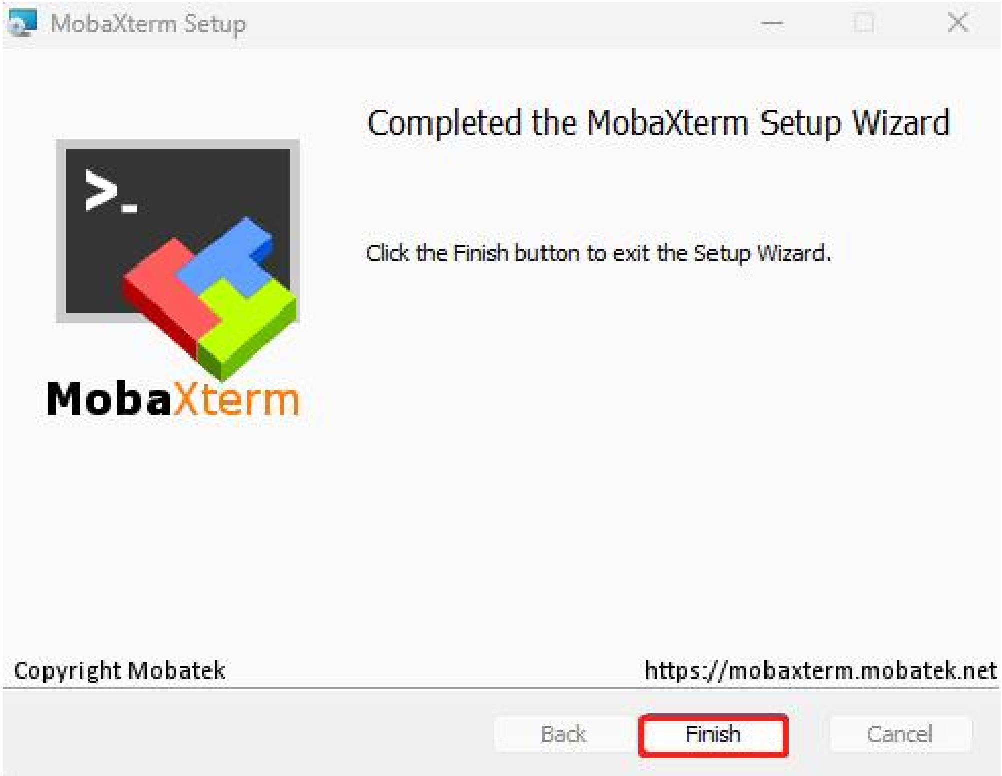


2) Follow the below pictures to finish the installation.




MobaXterm Setup
Destination Folder
Click Next to install to the default folder or click Change to choose another.
Install MobaXterm to:
C:\Program Files (x86)\Mobatek\MobaXterm\
Change...
Back
Next
Cancel

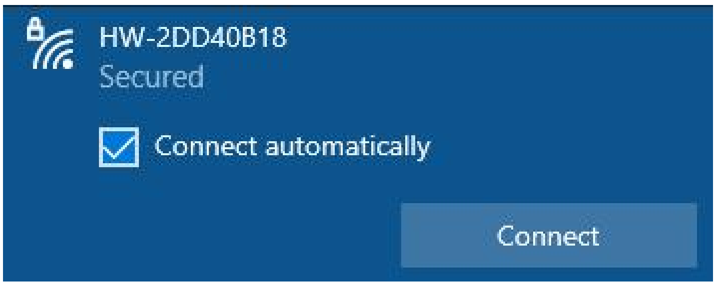

MobaXterm Setup
Ready to install MobaXterm
Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.
Back
Install
Cancel



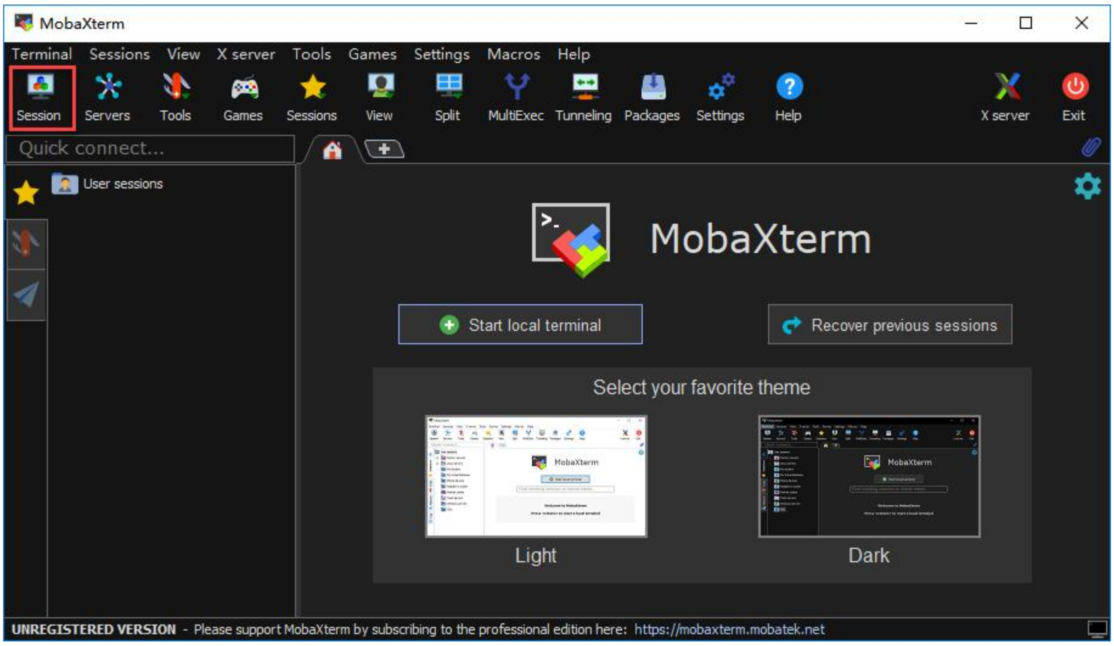
11.4 Connect Robot to MobaXterm

11.4.1 Connection

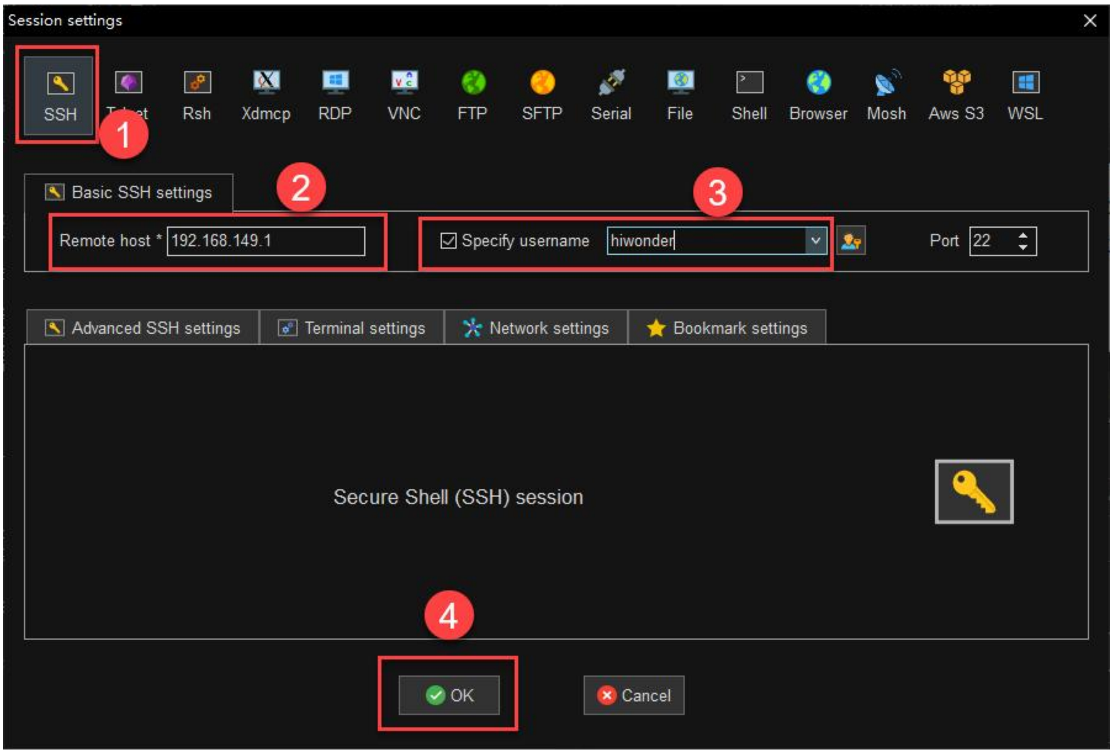
1) The default connection mode for the robot is AP direct connection mode. After the robot boots up successfully, it will generate a WiFi starting with the letter “**HW**”. Connect the computer to this WiFi.



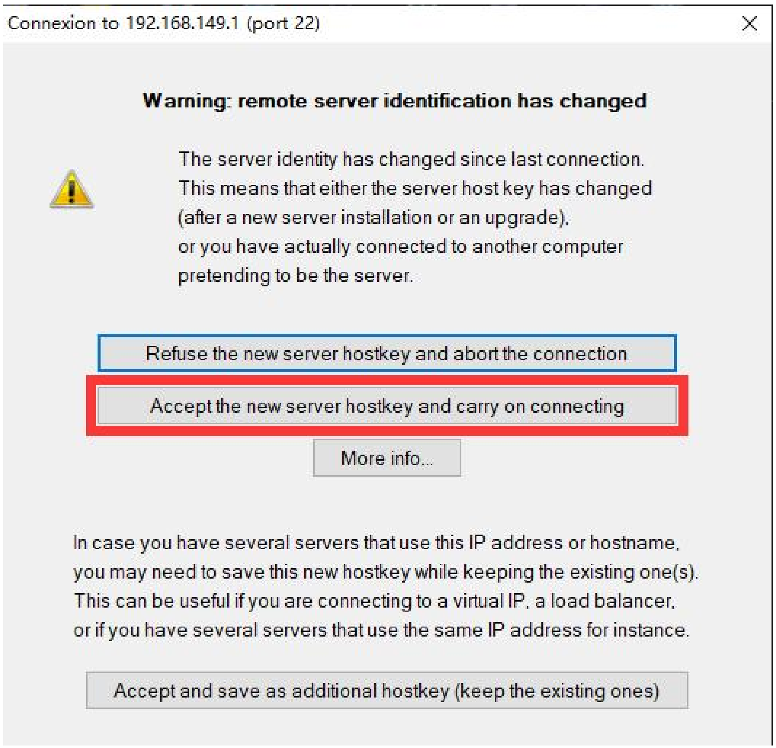
2) Open MobaXterm. Click-on ”**Session**” at the upper right corner to create a new session.



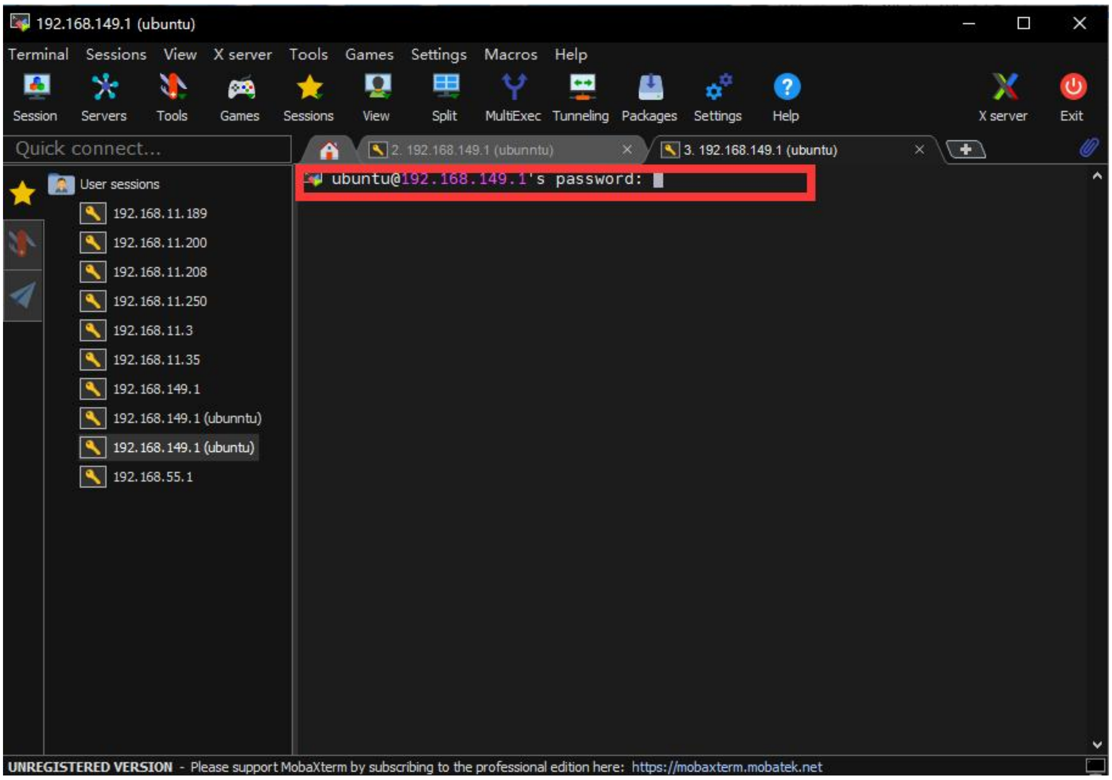
3) Select "**SSH**", then enter robot's IP. Take direct connection mode as example. Enter the IP "**192.168.149.1**", tick "**Specify username**", then enter the username "**hiwonder**". Lastly, click-on OK.



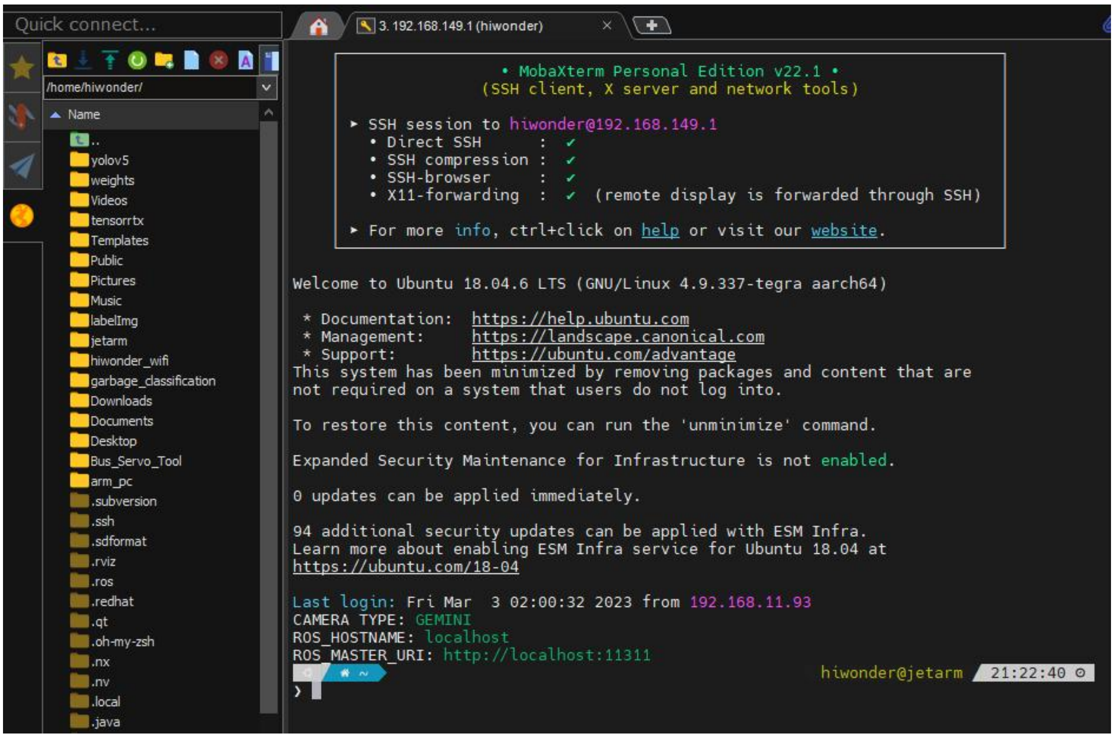
4) Select the option outlined with the red box.



5) The password is “**hiwonder**”. When you entering the password, the password will be hidden. Once you finish entering the password, press Enter key.



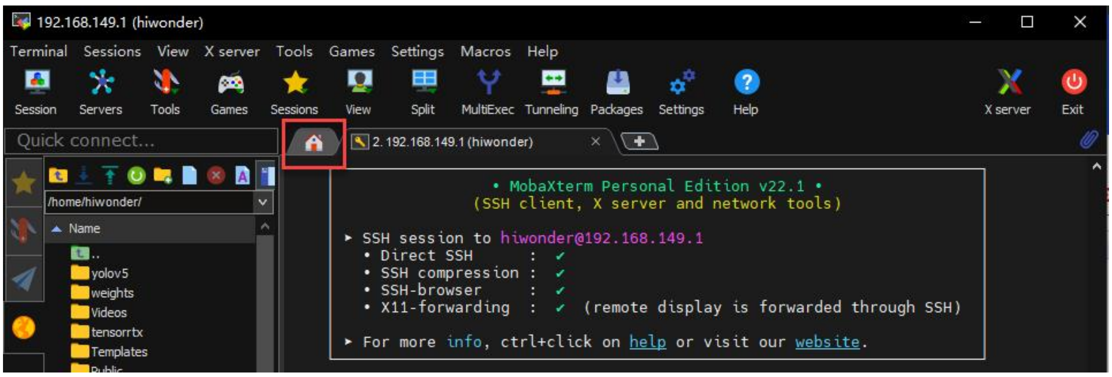
6) When you log in successfully to the system, the interface will appear as shown in the image.



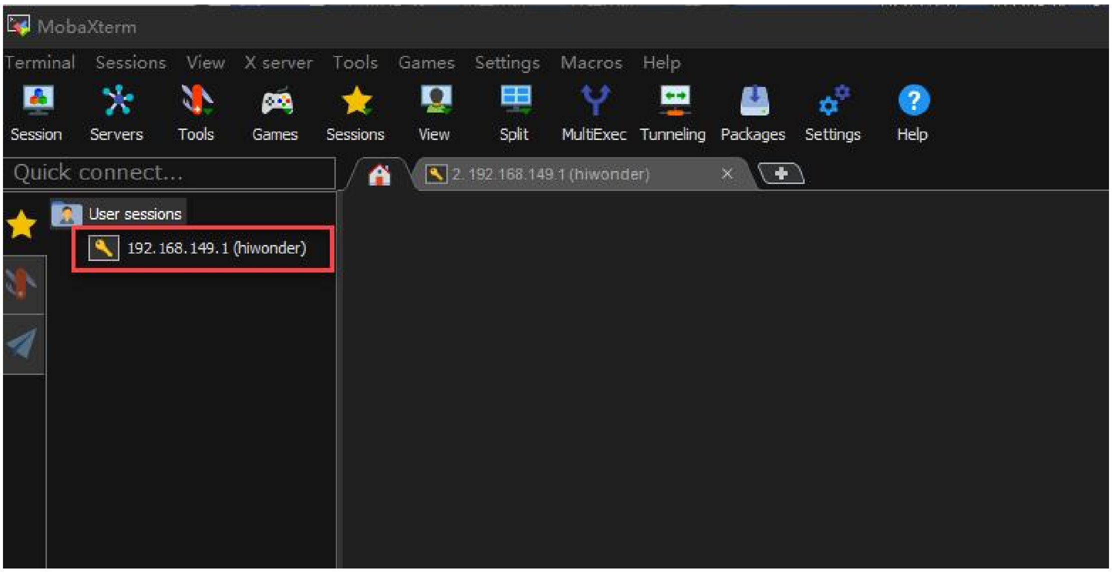
11.4.2 Open Multiple Dialogue Boxes

Each dialogue box can only run a single program command. When running a program, multiple commands for control may need simultaneously. In this case, it is necessary to open multiple dialogue boxes. The specific operations are as follow:

1) Click on .

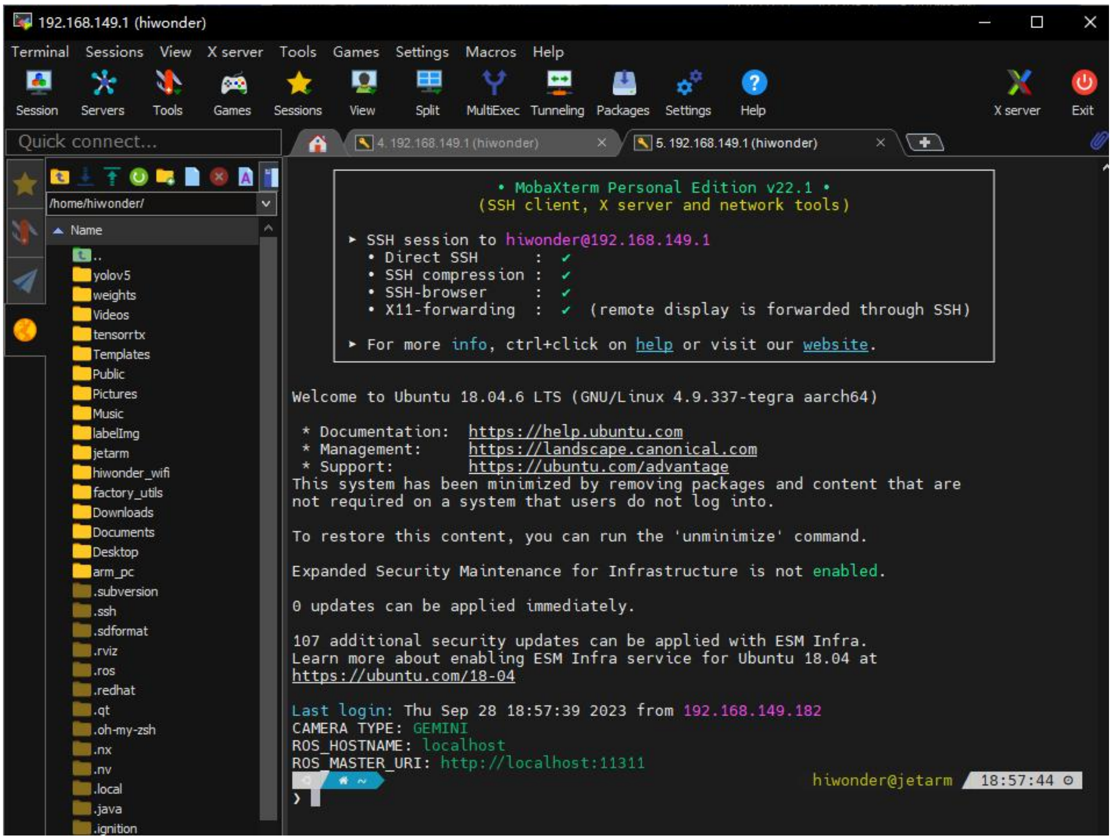


2) Double click the red box.



3) Enter the username "**hiwonder**" and press Enter.

4) Upon appearing the below information, the new dialogue box is opened successfully.



12. Position Adjustment for Object Gripping and Placing

Note:

1. Once the parameters adjusted take effect, the gripping and placing effects of color sorting, tag sorting, intelligent saocking (it can only be executed via app), and waste sorting.

2. Prior to shipment, the robotic arm has undergone calibration. However, discrepancies between the calibration environment and the actual usage environment, coupled with potential vibrations during transport, may result in suboptimal execution of sorting, stacking, and other games, causing the gripper to inaccurately pick up and place blocks.

3. In the event of the aforementioned issues, refer to the following instructions to adjust the gripping and placing positions of the robotic arm.

12.1 Preparation

1. Place and start the robotic arm on map. Connect it on app according to “5. APP Installation and Connection”.

2. Adjust the color threshold according to “9. Color Threshold Adjustment”.

3. Check whether the robotic arm has deviation. If there is a deviation, please adjust according to “4. 4 Deviation Adjustment”.

4. Calibrate the position according to “7. Position Calibration”.

5. Re-start the robot sorting or stacking games to check if there is any deviation during gripping and placing. If there is any deviation, then refer to the following instructions for adjustments (you may skip this section if the gripping and placing effects are good).